CSE4007
Artificial Intelligence

Lecture 13: Self-Supervised Learning

Department of Data Science
Hanyang University

~~Previously on Machine Learning~~ 1-2-3 Deep Learning in this course

- Collect data ~~and extract features~~

- Build model: choose hypothesis class $\mathcal{H}$ and loss function l
  - Recent deep learning:
    - Simultaneously learn feature extractors and classifiers/regressors
    - Caveat: early two-stage deep learning approaches are still sometimes used
- Optimization: minimize the empirical loss

Supervised Learning

Regression Task

- Linear Regression $f_{\boldsymbol{w}}(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{x}$

$$\mathcal{L}(f_{\boldsymbol{w}}) = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} - y^{(i)})$$

Classification Task: parametric approach

- Logistic Regression $\hat{y} = \sigma(w_0 + \mathbf{w}^T\mathbf{x})$

$$\mathcal{L}_{\mathrm{CE}}(\sigma(z), y) = y\log(1+e^{-z}) + (1-y)\log(1+e^{z})$$

- Softmax Classifier $\hat{\mathbf{y}} = \mathrm{softmax}(\mathbf{z})$

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{k=1}^{K} y_k \log \hat{y}_k$$

New tasks!

- Object Detection
  - Localization (regression) + Classification
- Segmentation
  - Per-pixel Classification

Recap: Supervised Learning

- Learn from data to achieve generalization:
  - To design algorithms that can be trained on data to make predictions for new examples/data
- Supervised Learning

Assume there is a mapping function:

$$\mathbf{x}^{(i)} \longrightarrow y^{(i)}$$

Train a model f that is as close to the mapping function as possible

- To make correct predictions for new examples

$$\mathbf{x}^{(i)} \longrightarrow \boxed{f} \longrightarrow \hat{y}^{(i)} \longleftrightarrow y^{(i)}$$

Problem: Supervised Learning is Expensive!

- Deep neural networks require a lot of data
- Supervised Learning requires labeled data
- Labeling (data annotation) takes a lot of time, human efforts, and $$

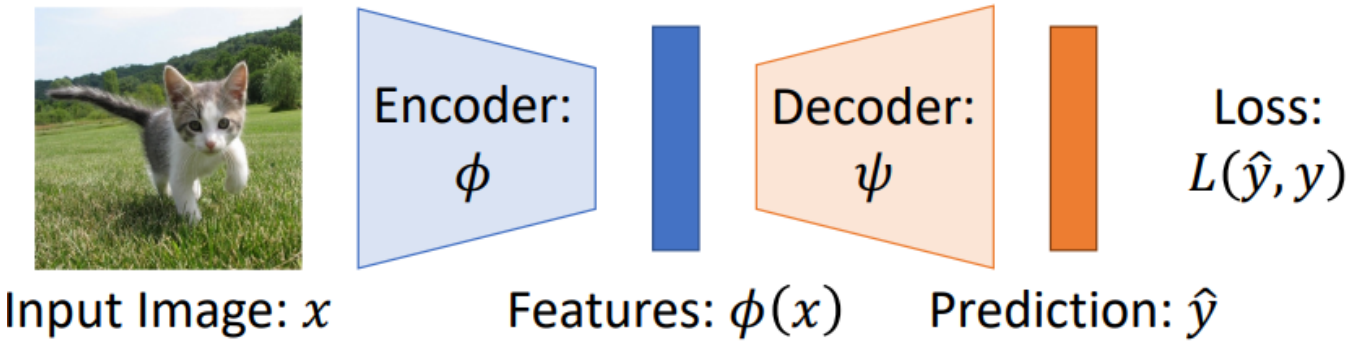
Solution: Self-Supervised Learning

- Today..
  - We will focus on methods that learn from “raw” data
  - Thus, no annotations required

- Unsupervised learning
  - No labels needed
  - Model isn’t told what to predict
- Semi-supervised learning
  - Trained with some labeled data and (a lot of) unlabeled data
- Self-supervised learning
  - No manual labels needed; but labels are created from the data itself
  - Trained to predict some naturally-occurring signal in the raw data rather than human annotations.

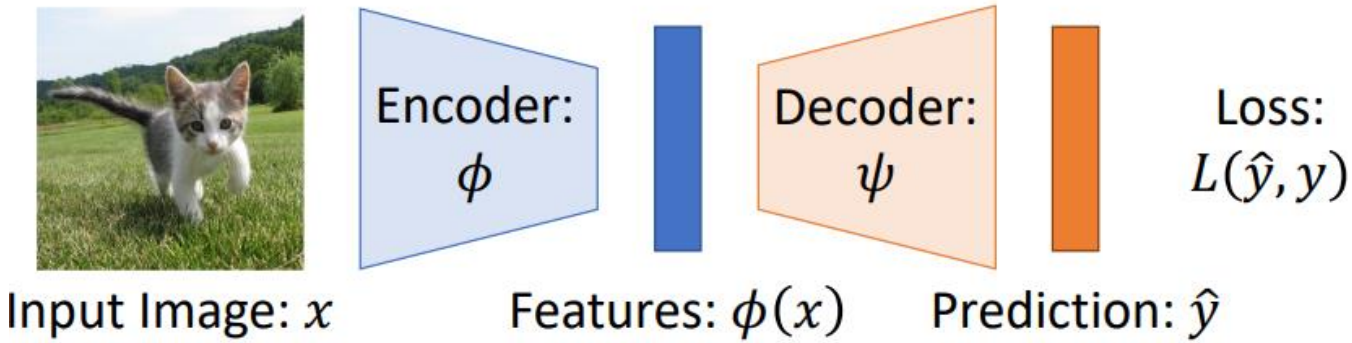
Self-Supervised Learning: Pretext

- **Step 1**: Pretrain a network on a pretext task that doesn't require supervision

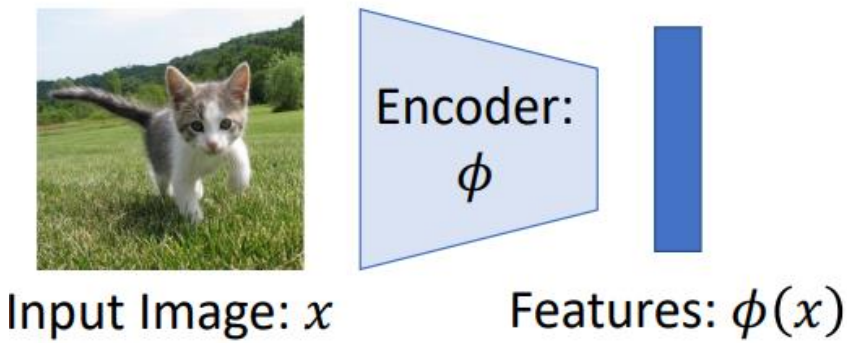


Self-Supervised Learning: Pretext then Transfer

- **Step 1**: Pretrain a network on a pretext task that doesn't require supervision



- **Step 2**: Transfer encoder to downstream tasks via linear classifiers, kNN, finetuning, etc.

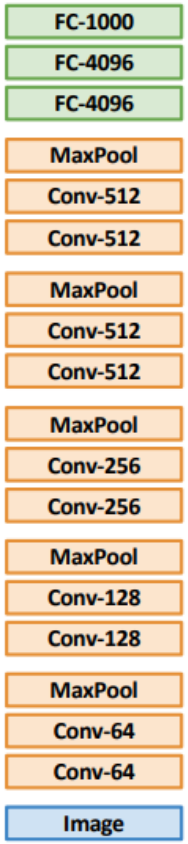


Downstream tasks: Image classification, object detection, semantic segmentation

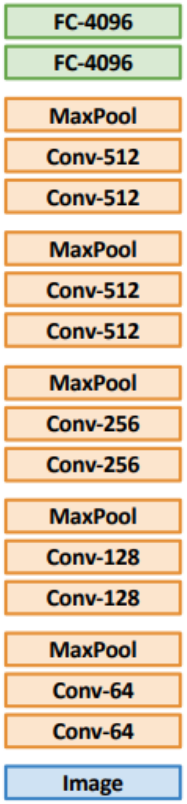
Ring a bell?

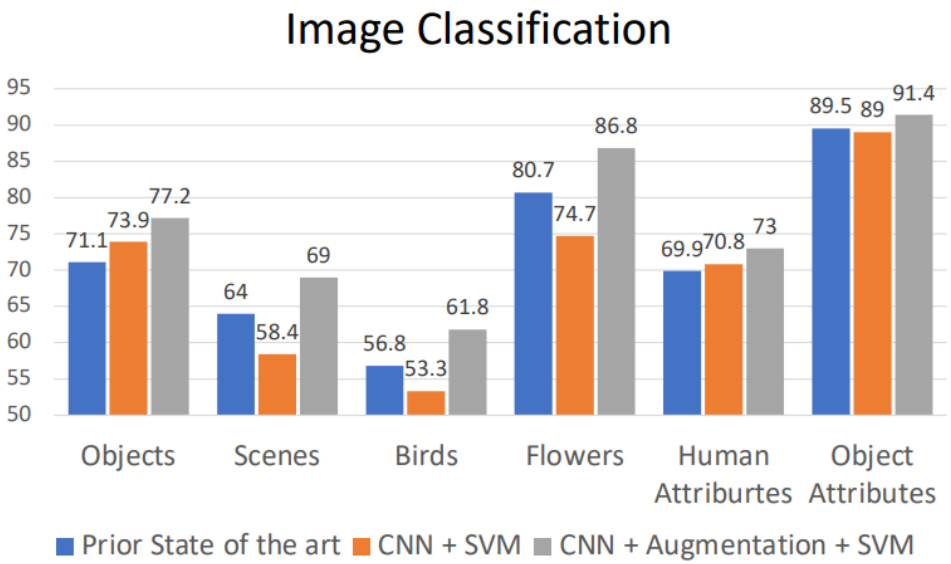
Recap - Transfer Learning: Feature Extraction

1. Train on ImageNet



2. Extract features with CNN, train linear model

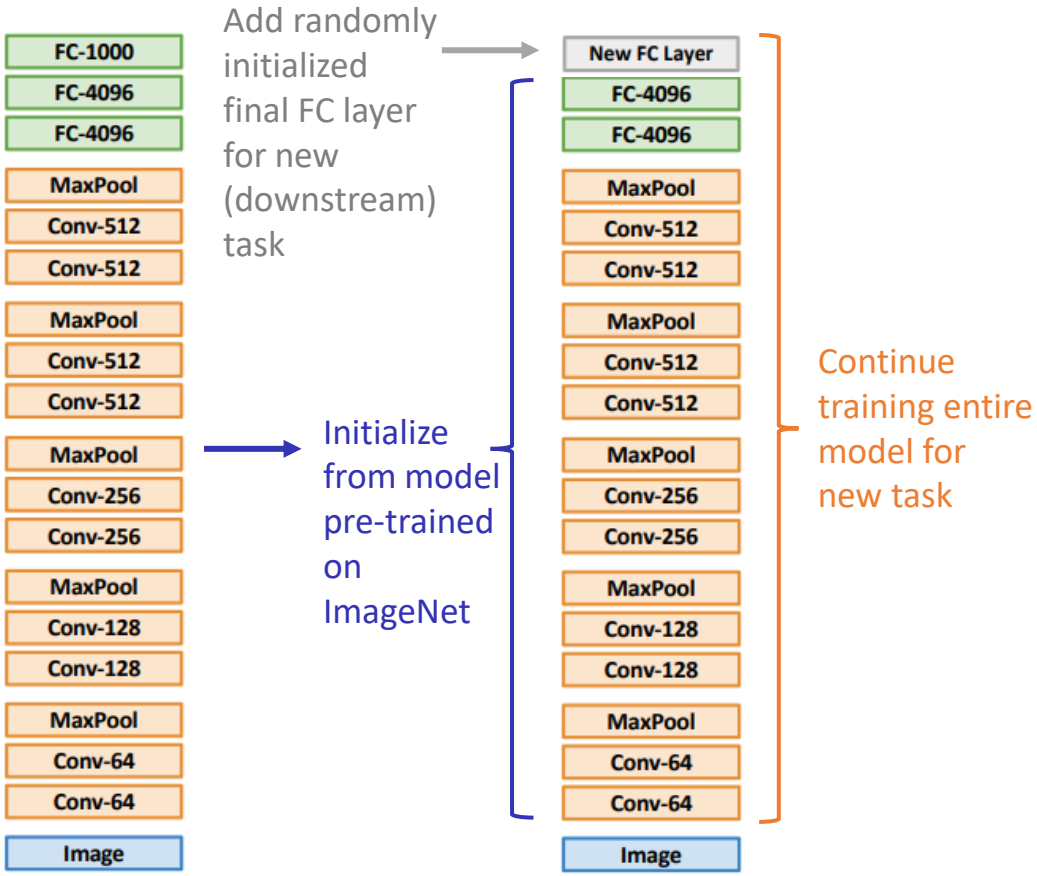




Razavian et al, "CNN Features Off-the-Shelf: An Astounding Baseline for Recognition", CVPR Workshops 2014

Recap - Transfer Learning: Fine-tuning

1. Train on ImageNet



Self-Supervised Learning: Pretext then Transfer

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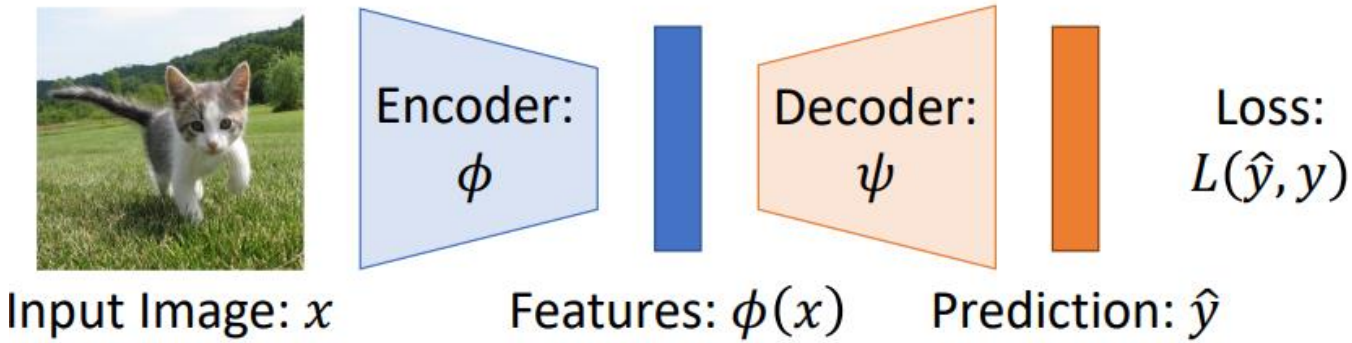

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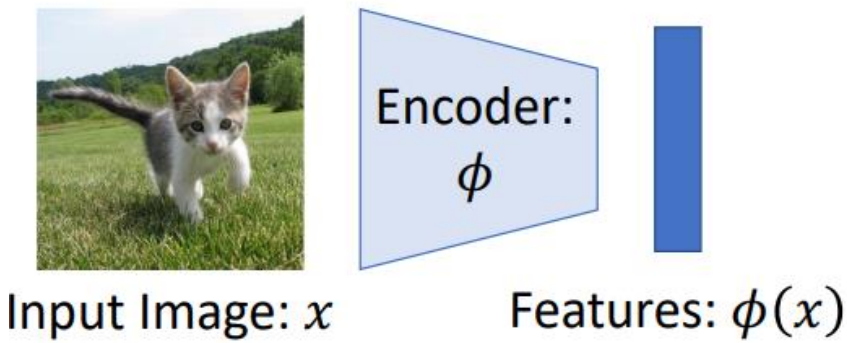


Downstream tasks: Image classification, object detection, semantic segmentation

Goal: Pretrain + Transfer does better than supervised pretraining, and better than directly training on downstream

We often call this pretraining-finetuning paradigm

Self-Supervised Learning: Pretext Tasks

Generative: Predict part of the input signal

- Autoencoders (sparse, denoising, masked)
- Autoregressive
- GANs
- Colorization
- Inpainting

Discriminative: Predict something about the input signal

- Context prediction
- Rotation
- Clustering
- Contrastive

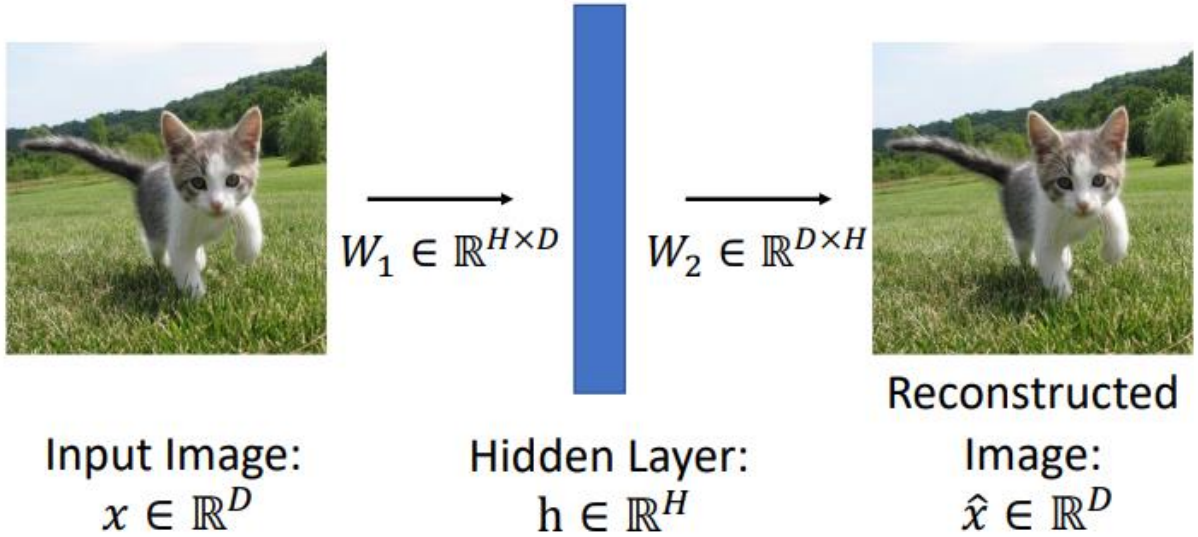
Multimodal: Use some additional signal in addition to RGB images

- Video
- 3D
- Sound
- Language

Recall: Autoencoder

Autoencoder tries to reconstruct inputs.
Hidden layer (hopefully) learns good representations.

<u>Generative</u> pretraining task!



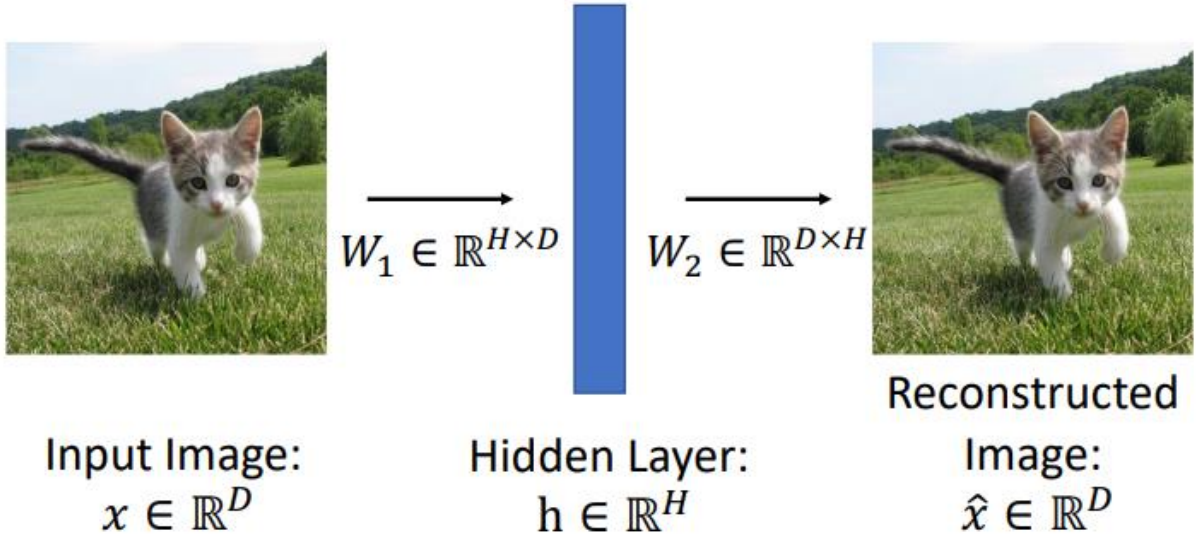
$$L(x) = R(x, \hat{x}) = \|x - \hat{x}\|_2^2$$

Recall: Autoencoder

Autoencoder tries to reconstruct inputs. Hidden layer (hopefully) learns good representations.

$H < D$ is the only thing forcing non-trivial hidden representations…

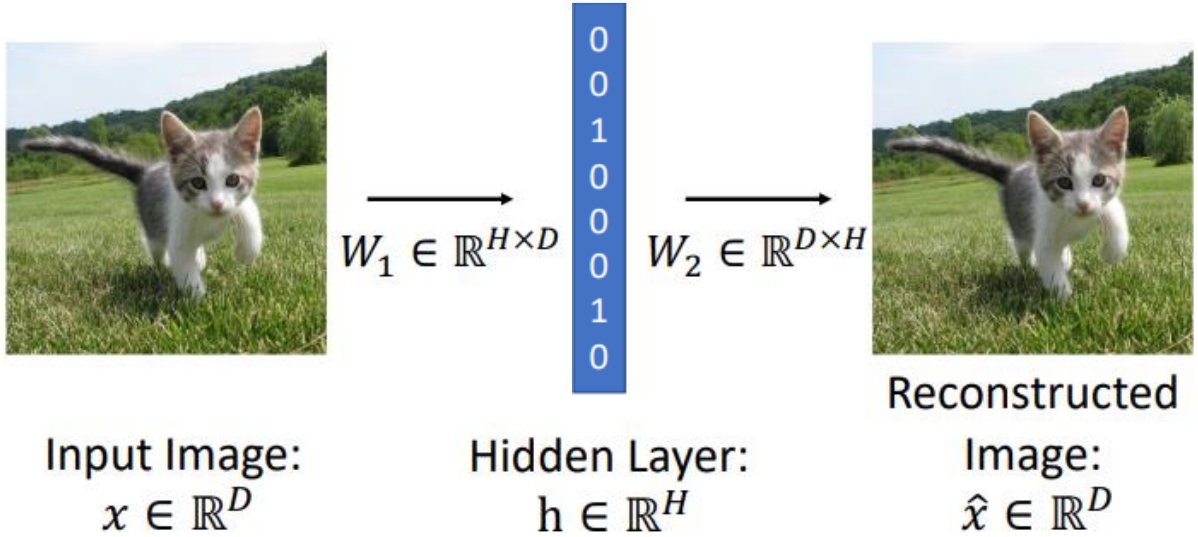
<u>Generative</u> pretraining task!



$$L(x) = R(x, \hat{x}) = \|x - \hat{x}\|_2^2$$

Sparse Autoencoder

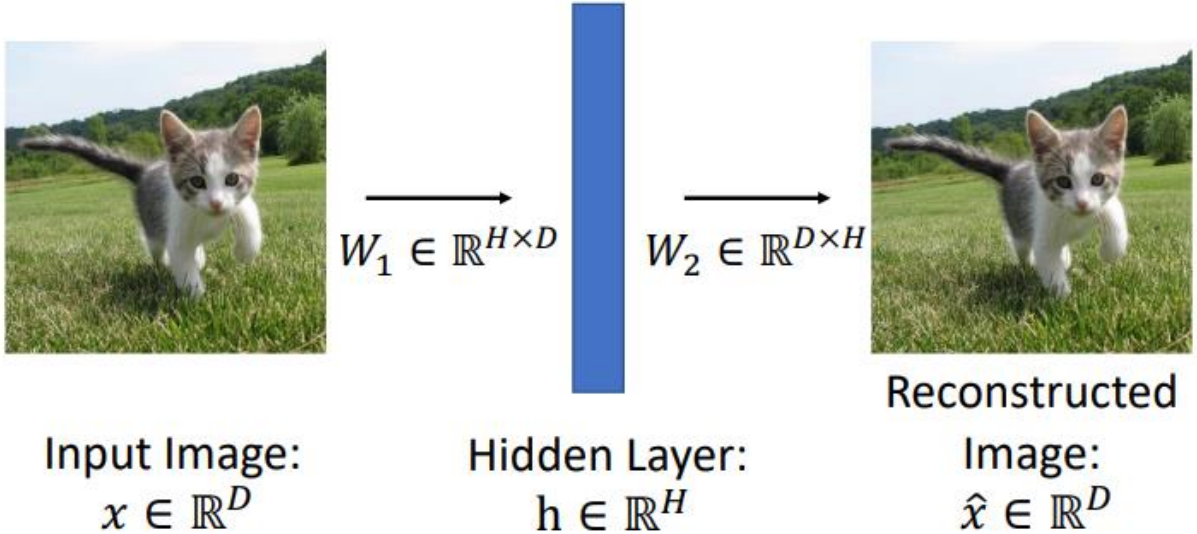
Train an autoencoder to **reconstruct inputs** with **sparse activations**. Many ways to implement sparsity penalties!



$$L(x) = R(x, \hat{x}) + \lambda S(h)$$
$$= \|x - \hat{x}\|_2^2 + \lambda \|h\|_1$$

Denoising Autoencoder

Train an autoencoder to reconstruct noisy inputs (pixels randomly set to zero).



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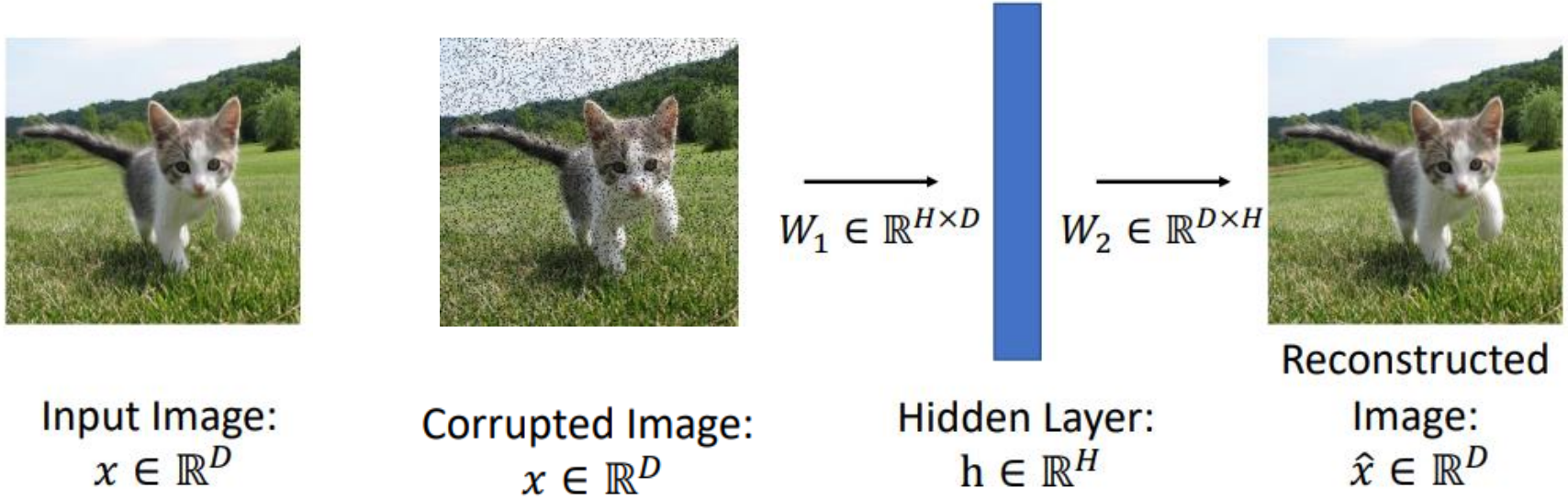


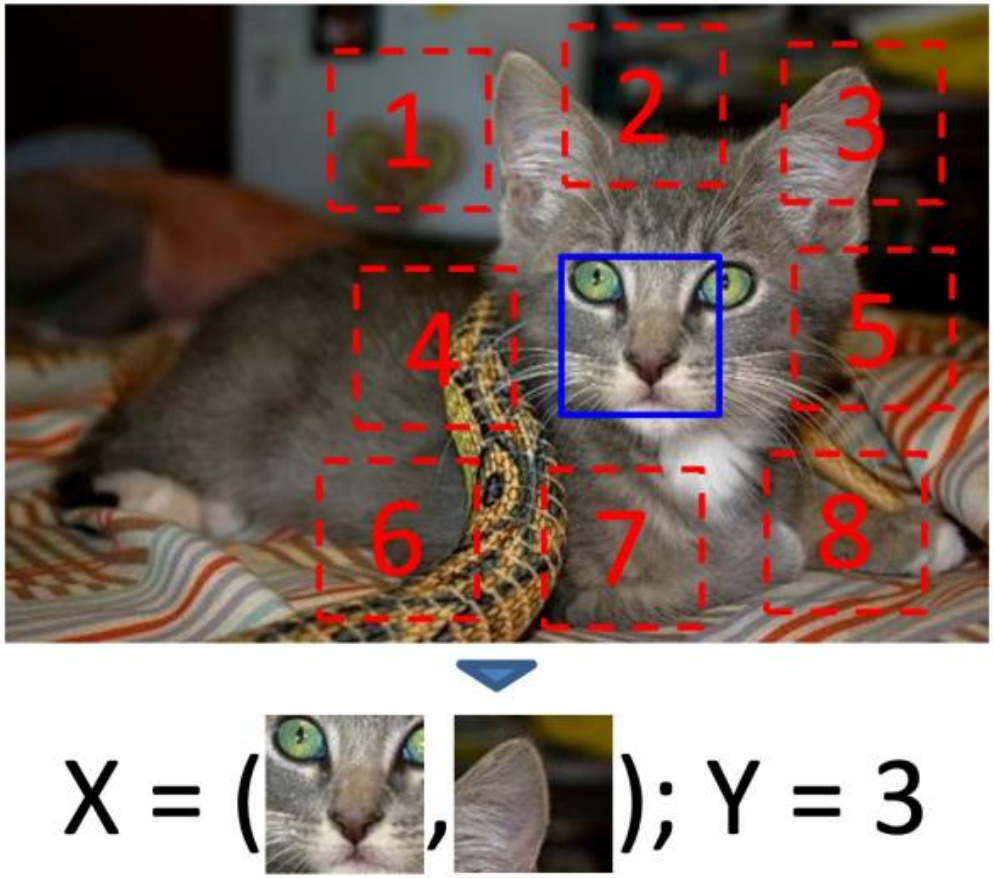
$$L(x) = R(x, \hat{x}) = \|x - \hat{x}\|_2^2$$

Context Prediction

Model is trained to predict the relative location of two patches from the same image.

<u>Discriminative</u> pretraining task

Intuition: Requires understanding objects and their parts



Doersch et al, “Unsupervised Visual Representation Learning by Context Prediction”, ICCV 2015

Context Prediction

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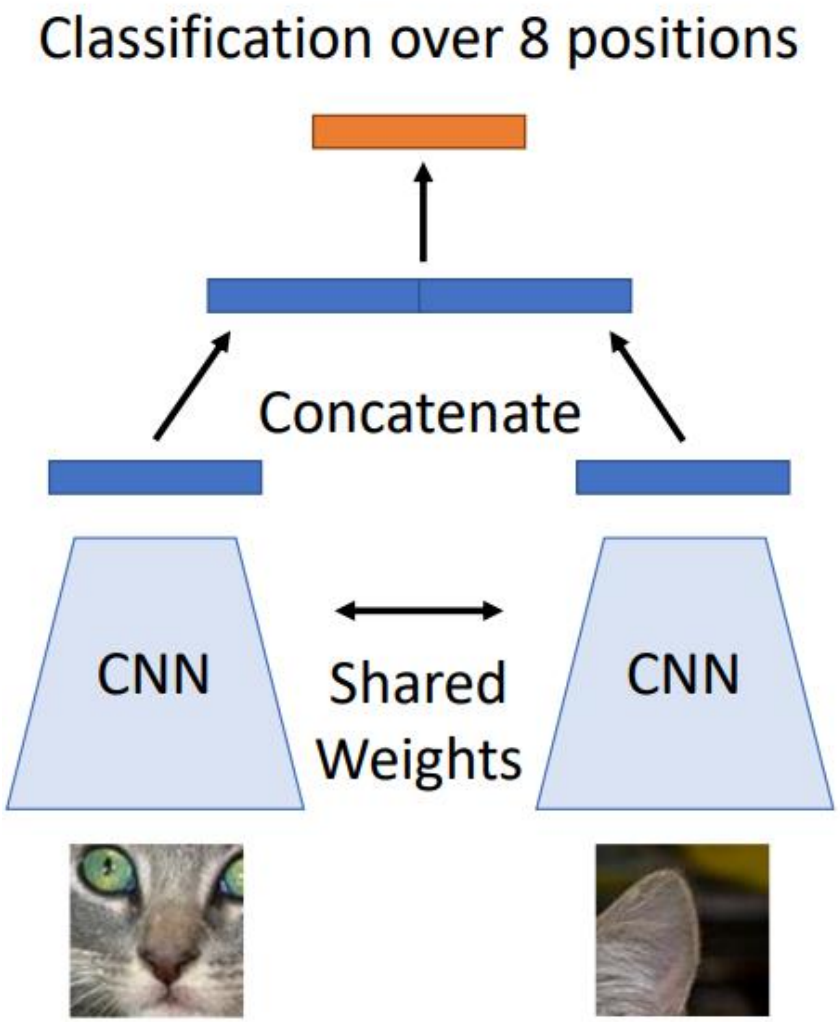
<u>Discriminative</u> pretraining task

Intuition: Requires understanding objects and their parts

Two networks with shared weights are sometimes called a "Siamese network"

-- Not really used these days



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Context Prediction: Nearest Neighbors in Feature Space

Input Patch

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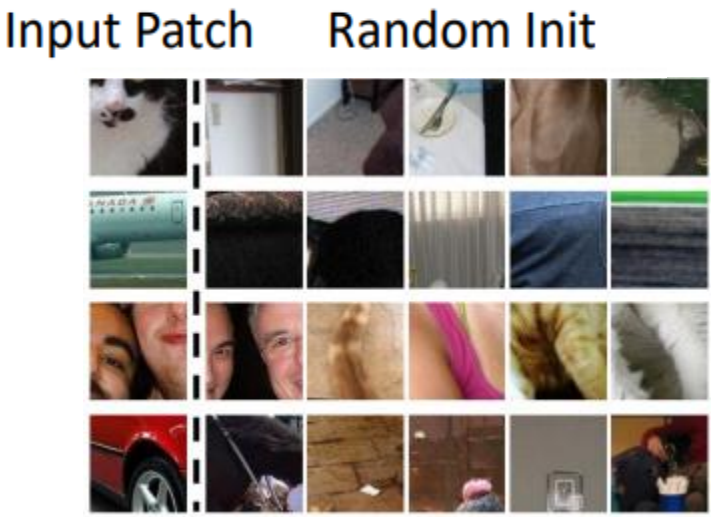
Context Prediction: Nearest Neighbors in Feature Space



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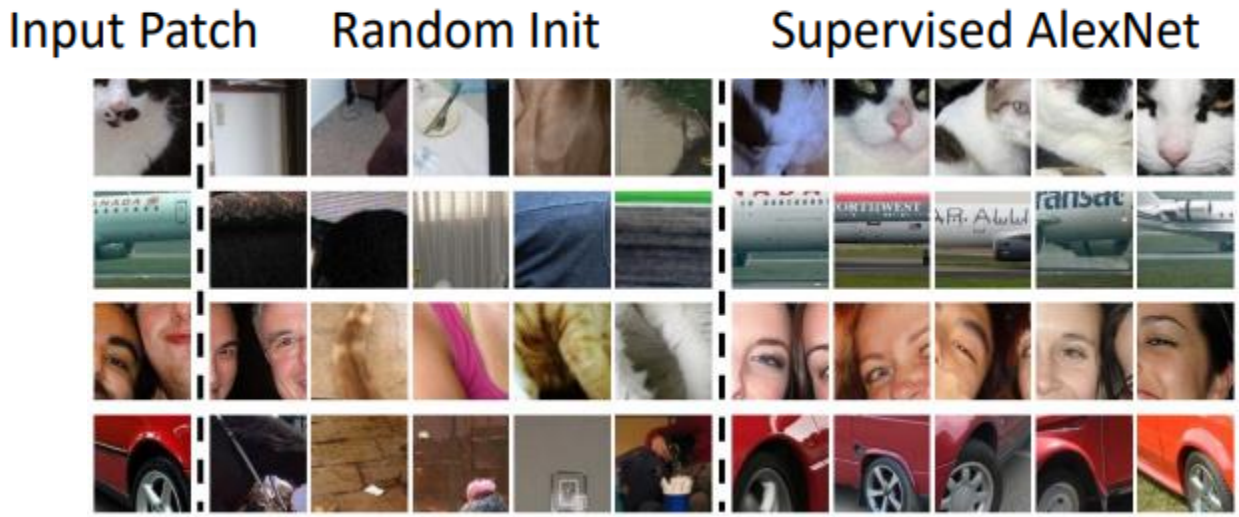

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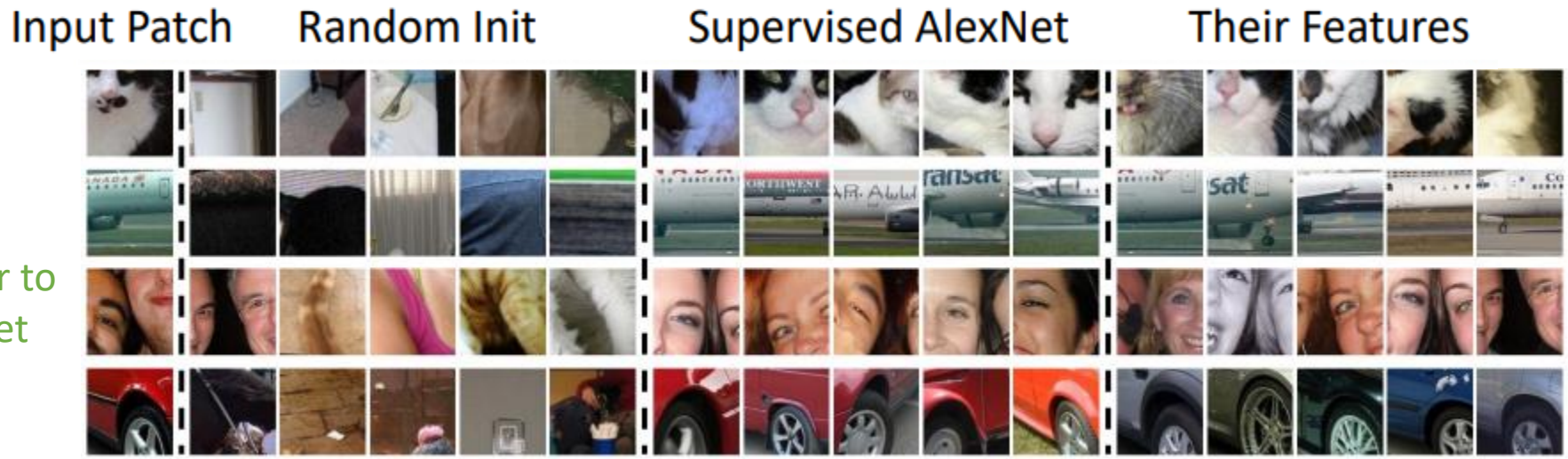
Context Prediction: Nearest Neighbors in Feature Space

Works well! Similar to AlexNet



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Context Prediction: Nearest Neighbors in Feature Space

Input Patch | Random Init | Supervised AlexNet | Their Features

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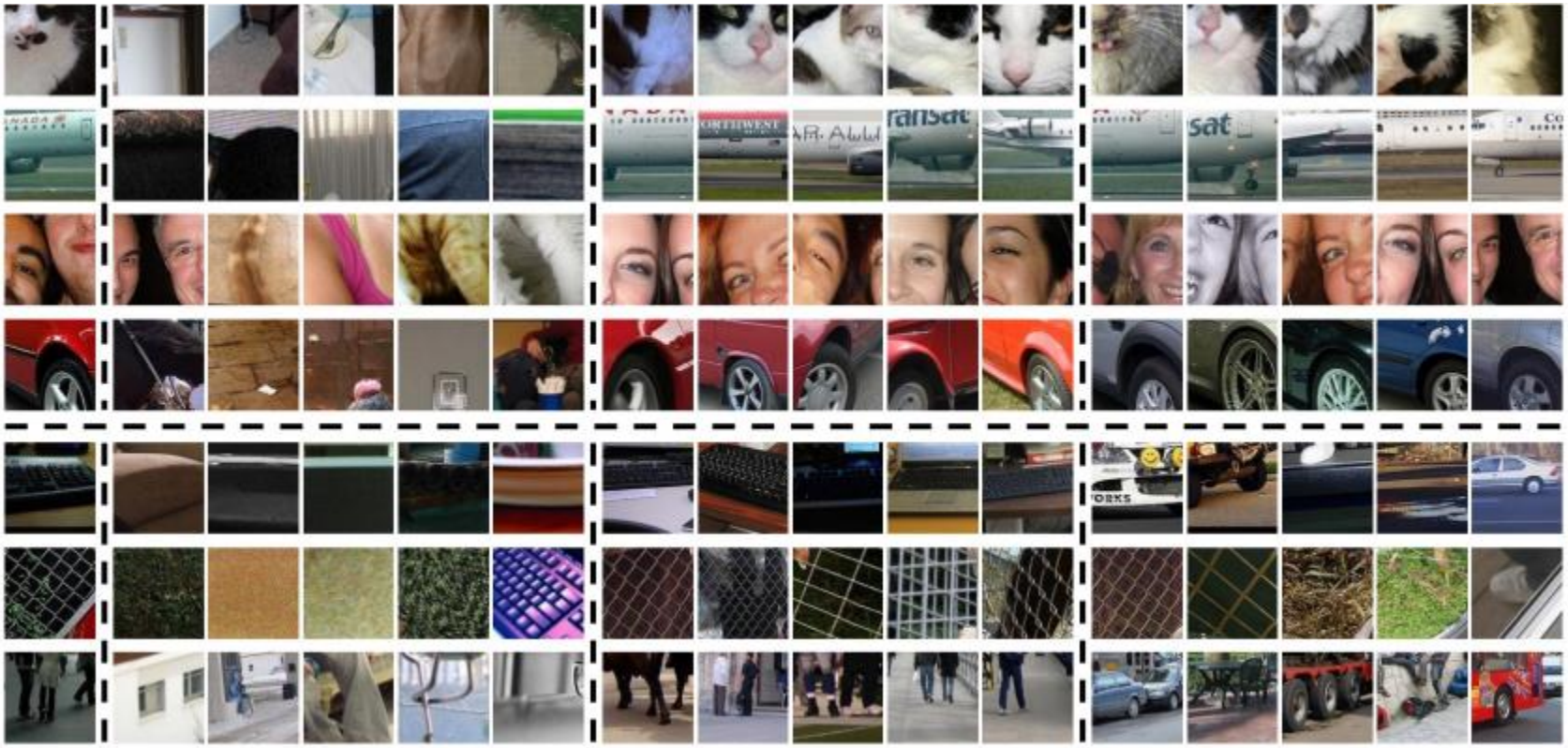

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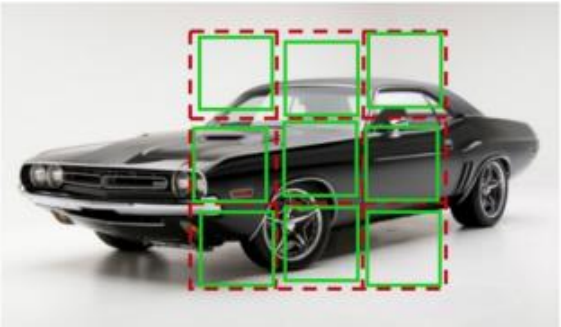
Extension: Solving Jigsaw Puzzles

Rather than predicting the relative position of two patches,
Predict permutation to “unscramble” 9 shuffled patches



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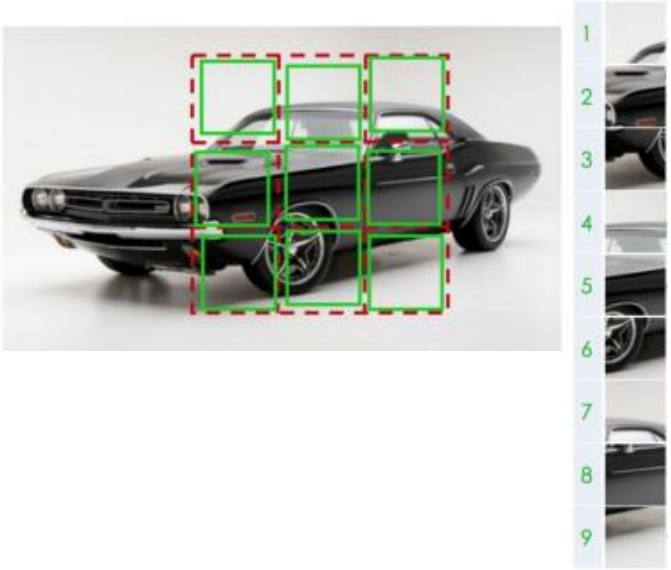

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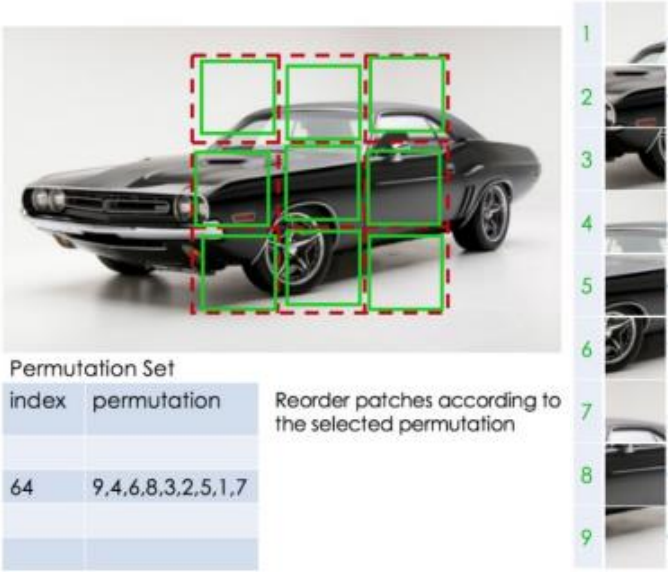

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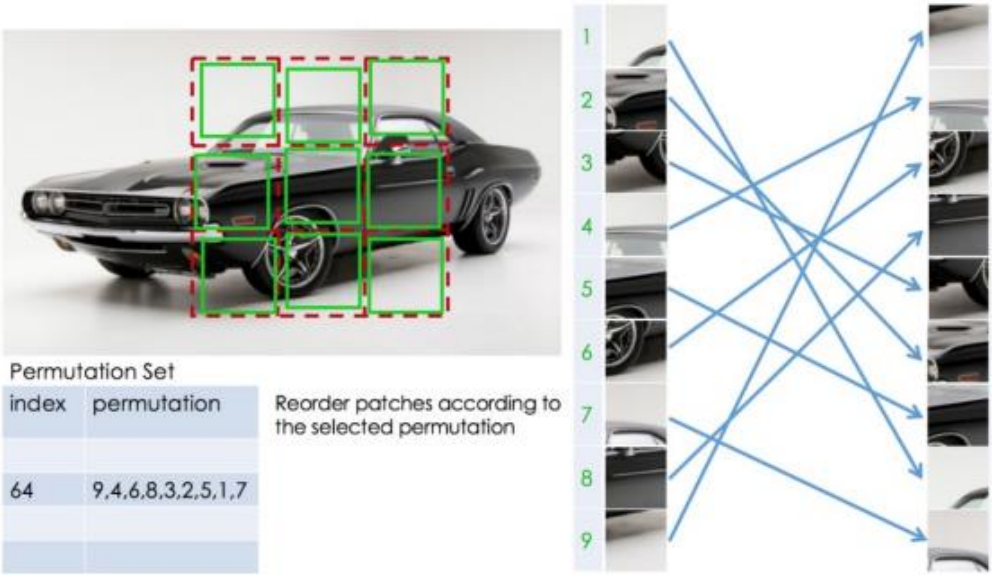

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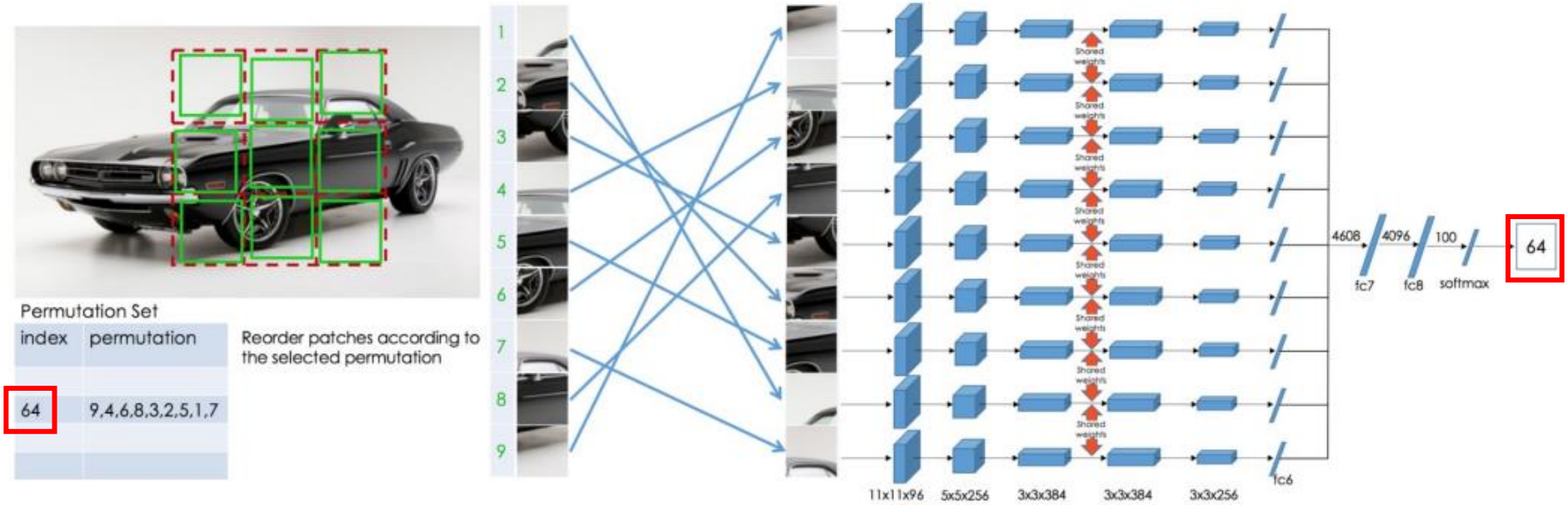

P-way classification: Predicting the permutation index.
P is the number of permutations used (P=100 in the original paper).

Noroozi and Favoro, "Unsupervised learning of visual representations by solving jigsaw puzzles", ECCV 2016

Extension: Solving Jigsaw Puzzles

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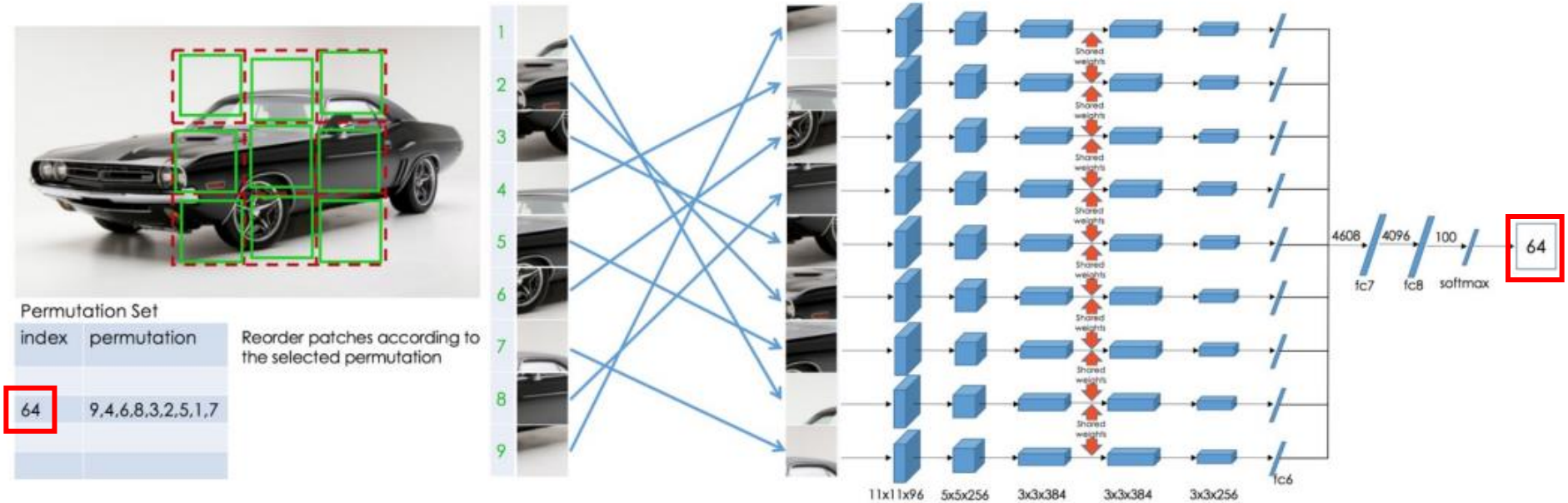
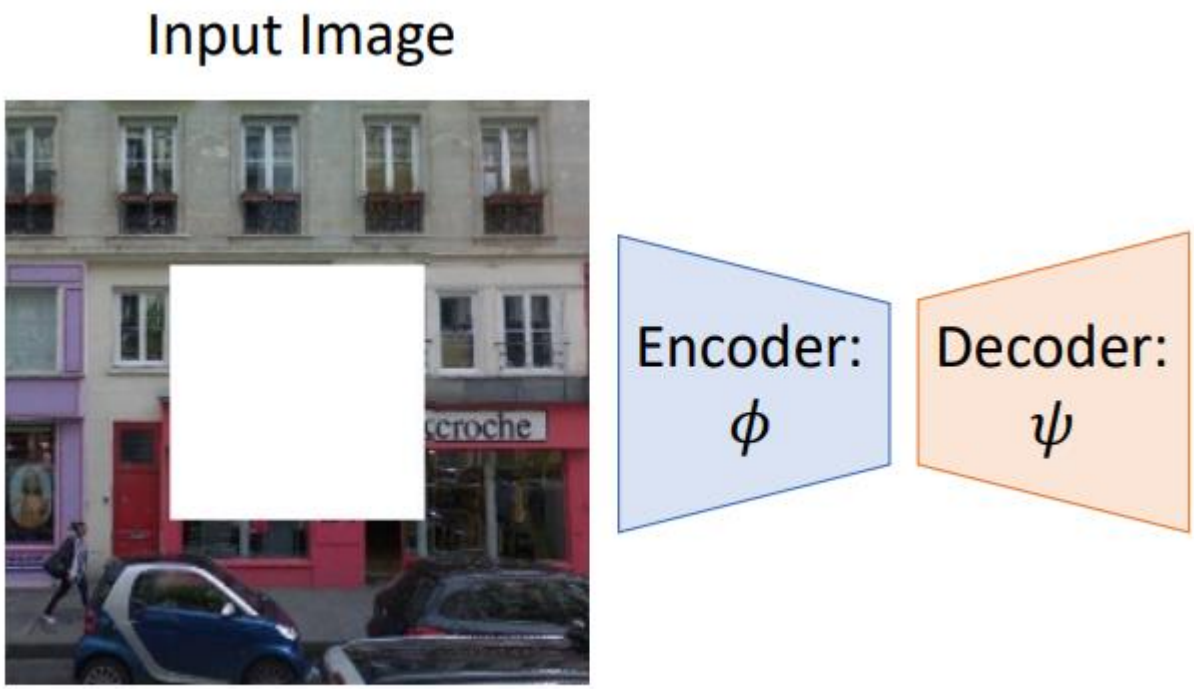


Problem: These methods only work on patches, not whole images!

Noroozi and Favoro, "Unsupervised learning of visual representations by solving jigsaw puzzles", ECCV 2016

Context Encoders: Learning by Inpainting



Pathak et al, “Context Encoders: Feature Learning by Inpainting”, CVPR 2016

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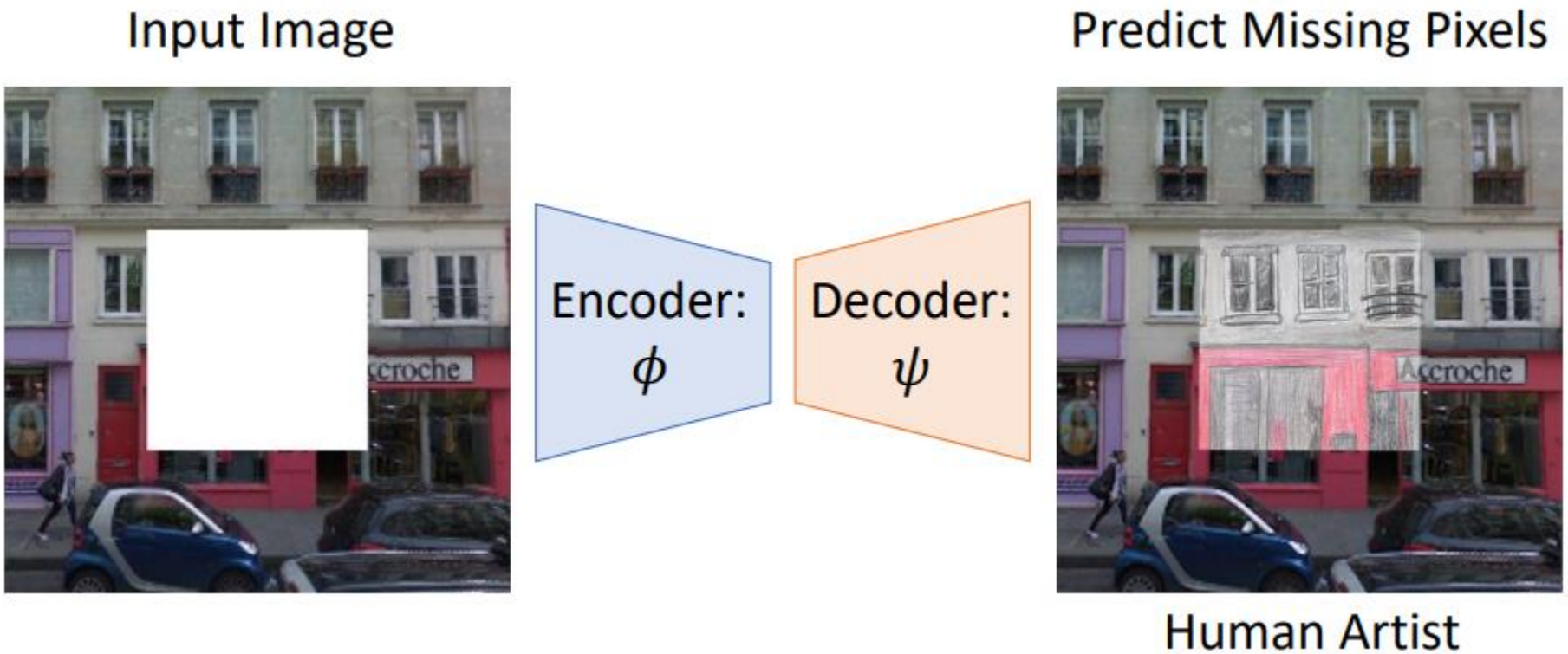

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Context Encoders: Learning by Inpainting

Input Image

Encoder: ϕ

Decoder: ψ

Predict Missing Pixels

L2 Loss
(Best for feature learning)

Pathak et al, "Context Encoders: Feature Learning by Inpainting", CVPR 2016

Context Encoders: Learning by Inpainting

Input Image

Encoder: ϕ

Decoder: ψ

Predict Missing Pixels

L2 + Adversarial Loss
(Best for nice images)

Pathak et al, "Context Encoders: Feature Learning by Inpainting", CVPR 2016

Colorization

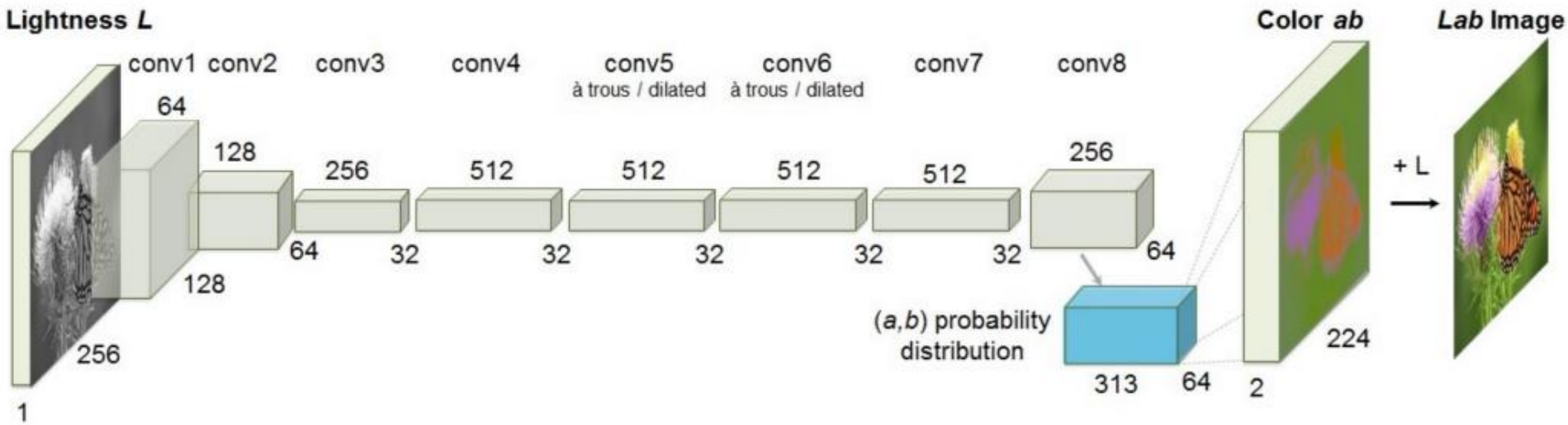
Intuition: A model must be able to identify objects to be able to colorize them

Input: Grayscale Image

Output: Color Image

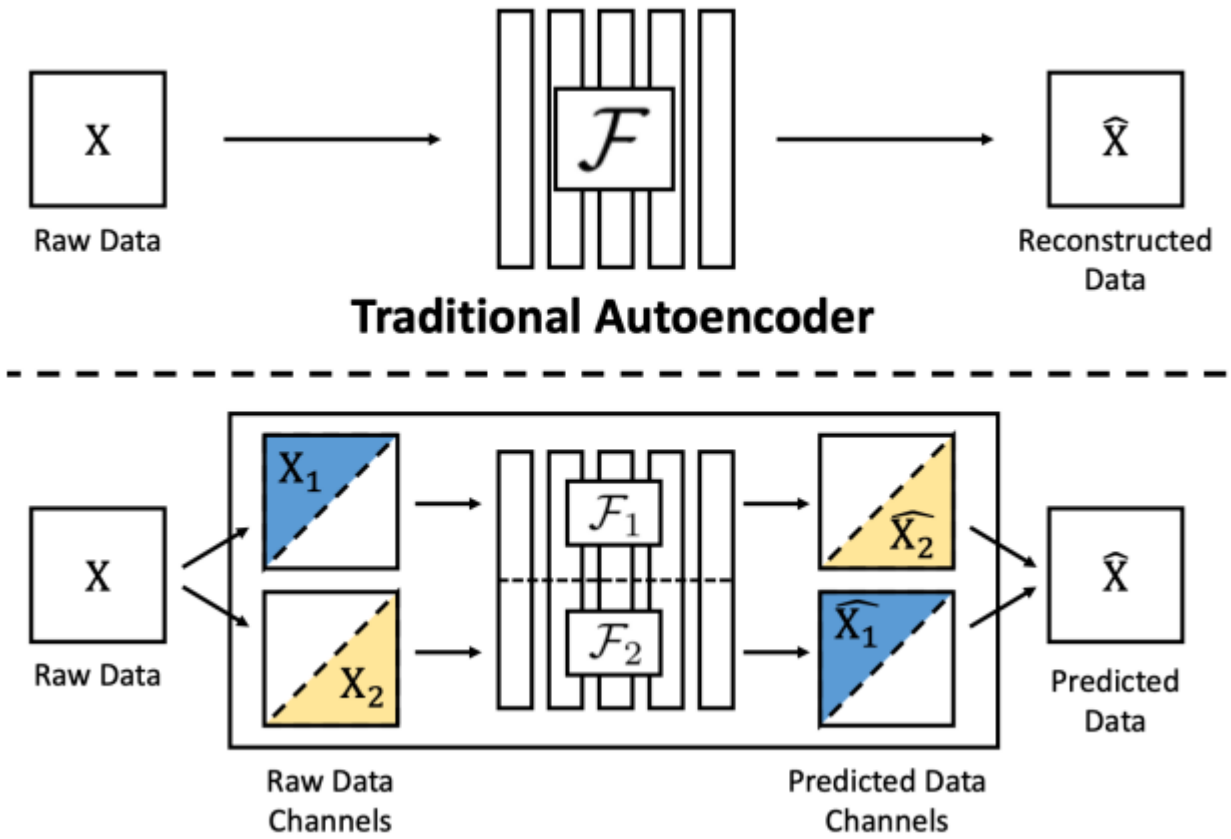
Zhang et al, “Colorful Image Colorization”, ECCV 2016

Colorization



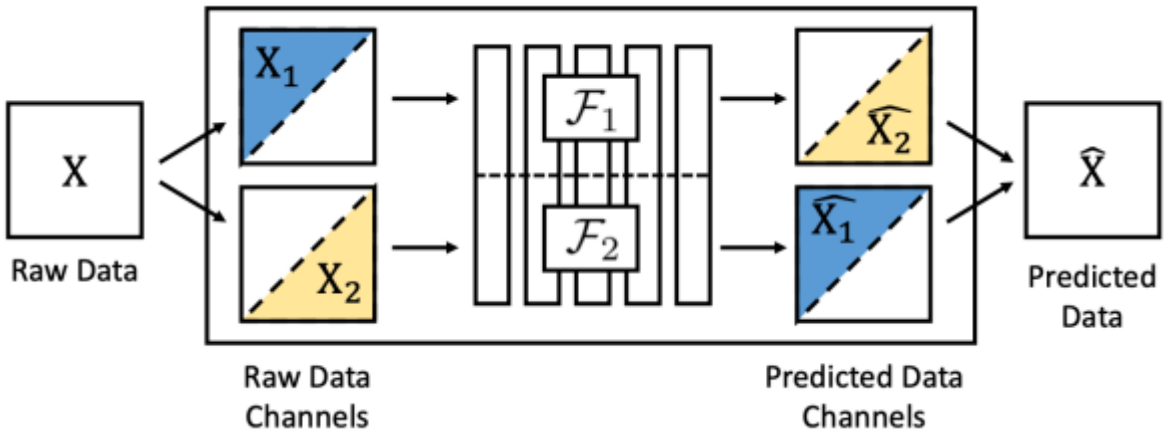
Zhang et al, “Colorful Image Colorization”, ECCV 2016

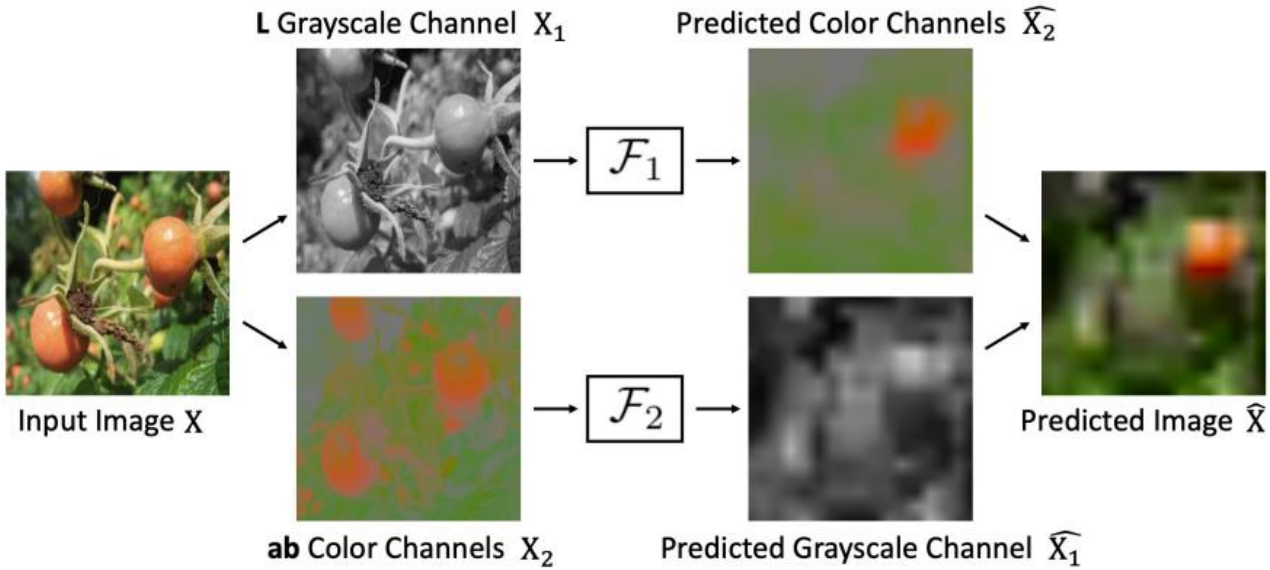
Colorization Extension: Split-Brain Autoencoder



Zhang et al, "Split-Brain Autoencoders: Unsupervised Learning by Cross-Channel Prediction", CVPR 2017

Colorization Extension: Split-Brain Autoencoder





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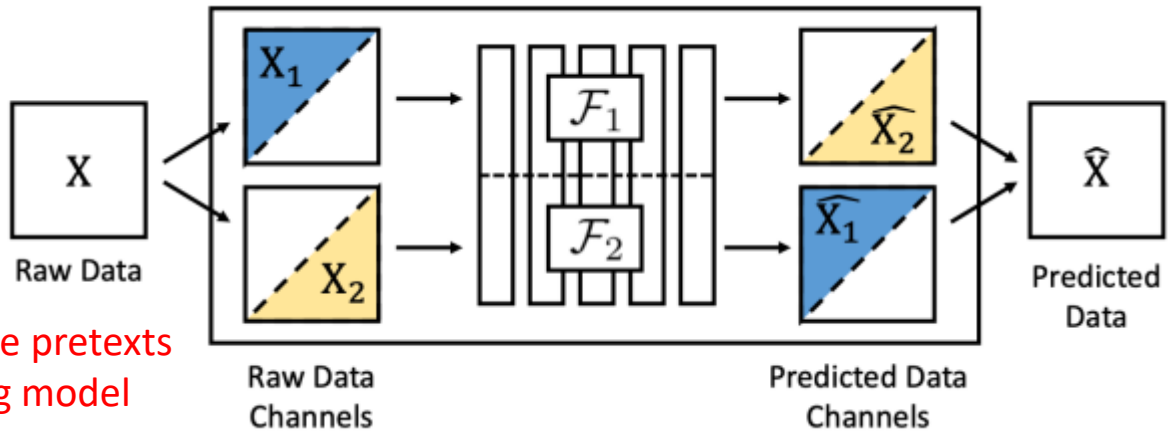
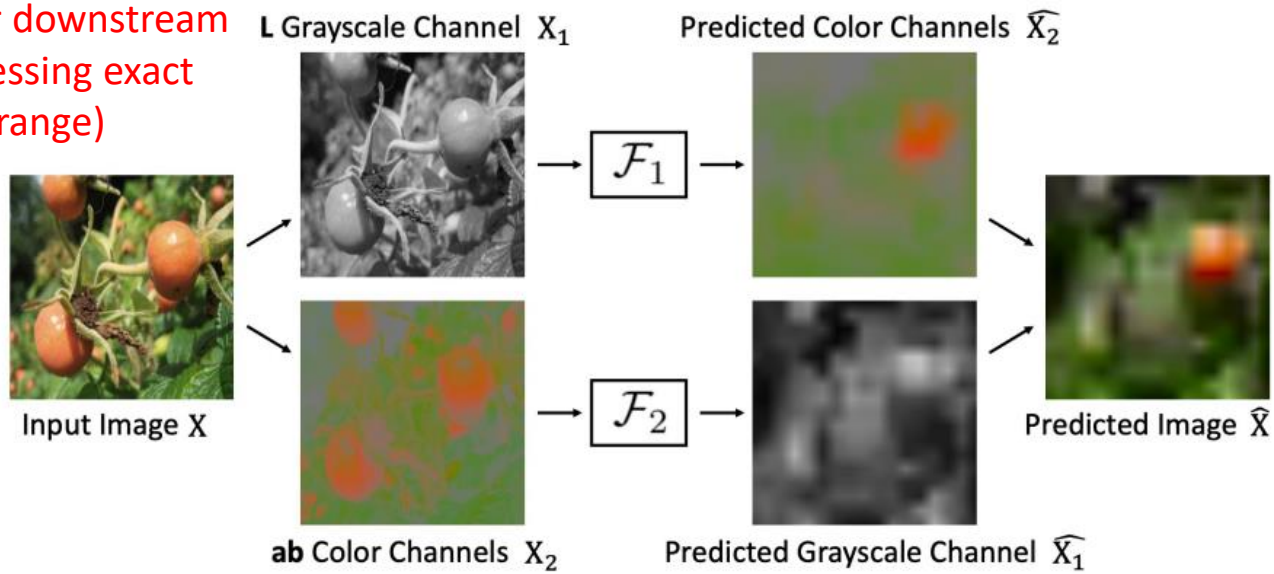


Concern: Generative pretexts encourage spending model capacity on details unimportant for downstream tasks (e.g., regressing exact right shade of orange)



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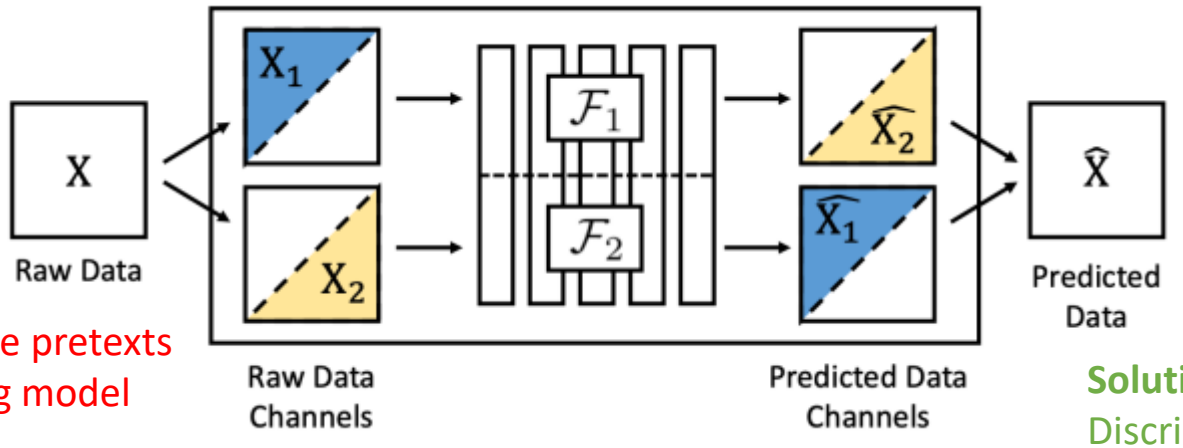
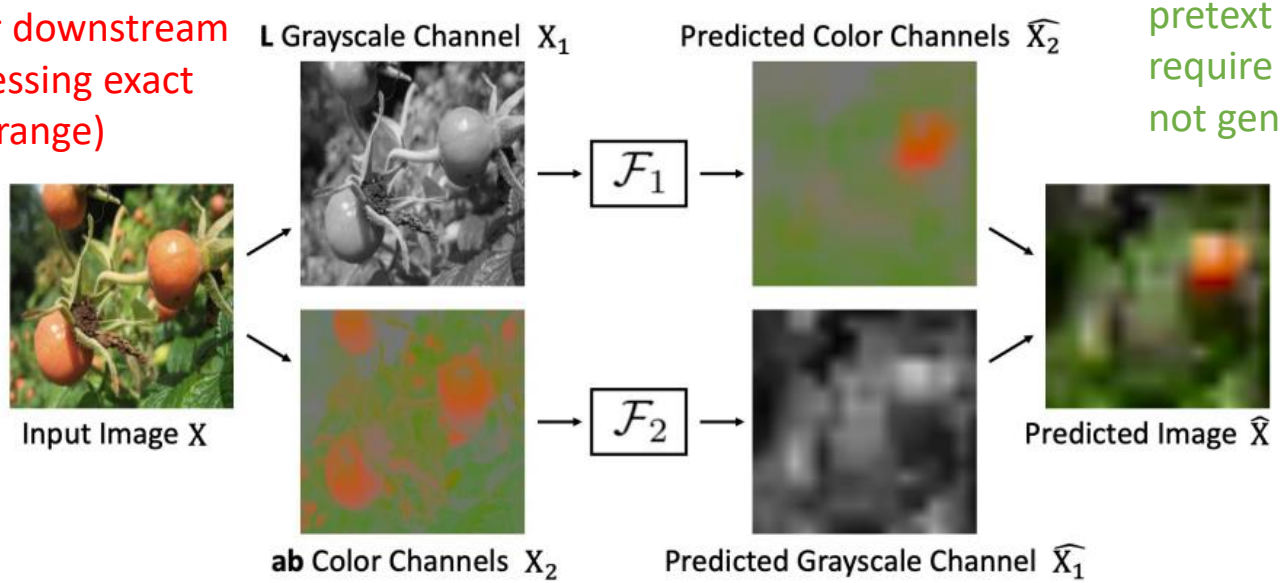


Concern: Generative pretexts encourage spending model capacity on details unimportant for downstream tasks (e.g., regressing exact right shade of orange)

Solution: Discriminative pretext tasks that require classification, not generation



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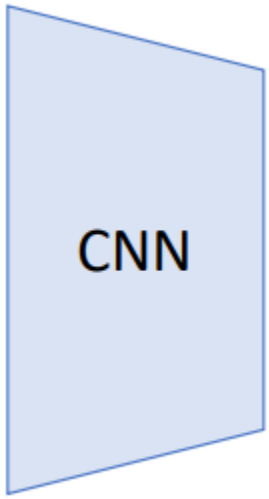
Deep Clustering

(1) Randomly initialize a CNN



Zhang et al, “Split-Brain Autoencoders: Unsupervised Learning by Cross-Channel Prediction”, CVPR 2017

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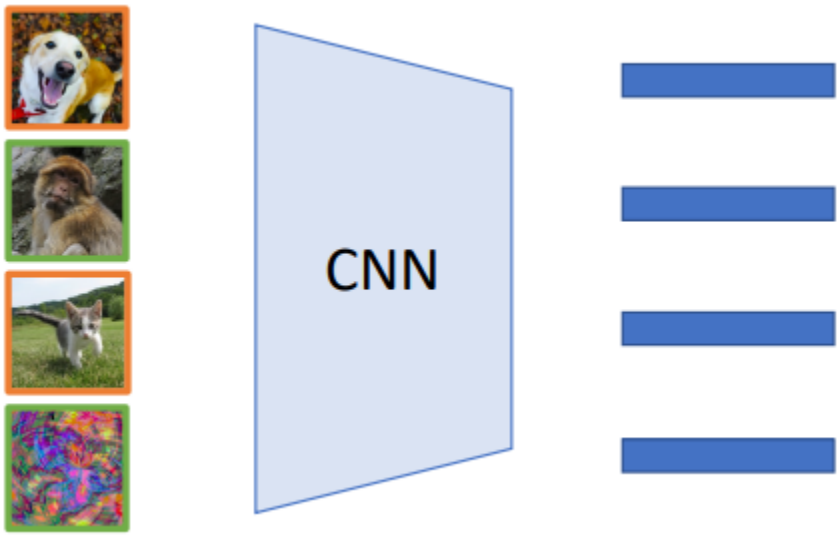

(2) Run many images through CNN,
get their final-layer features

Caron et al, “Deep Clustering for Unsupervised Learning of Visual Features”, ECCV 2018
Caron et al, “Unsupervised Pre-Training of Image Features on Non-Curated Data”, ICCV 2019
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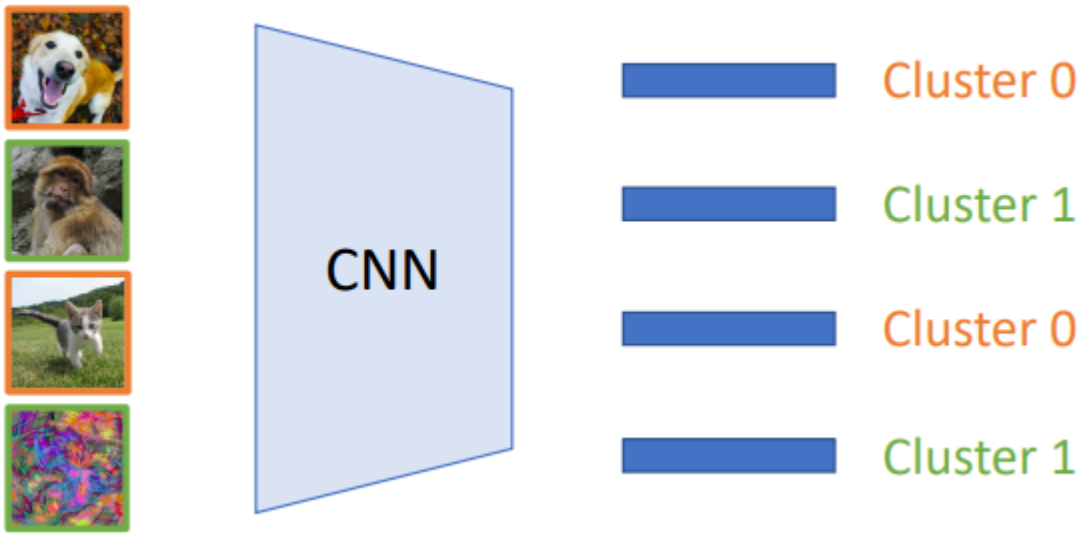

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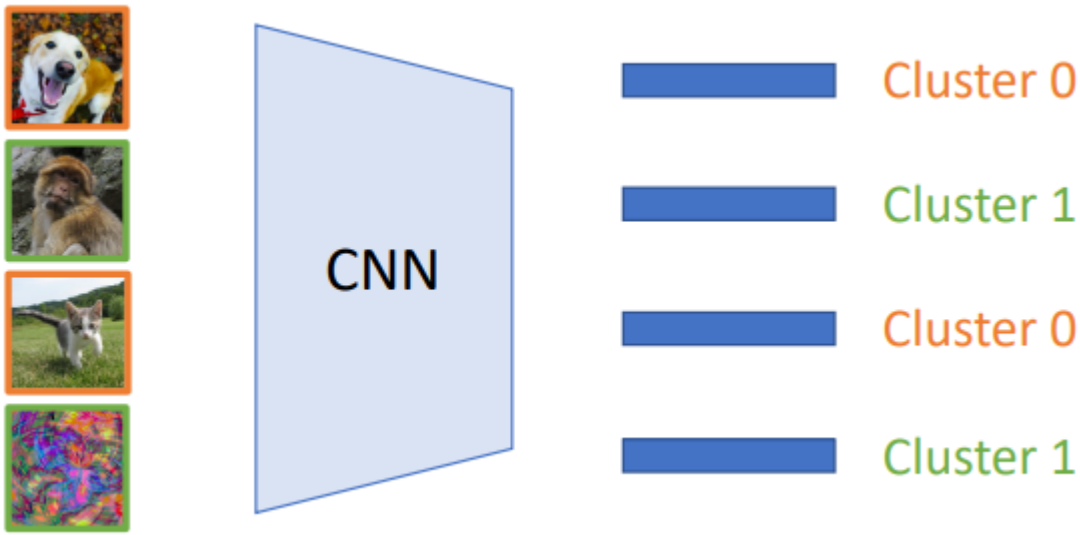

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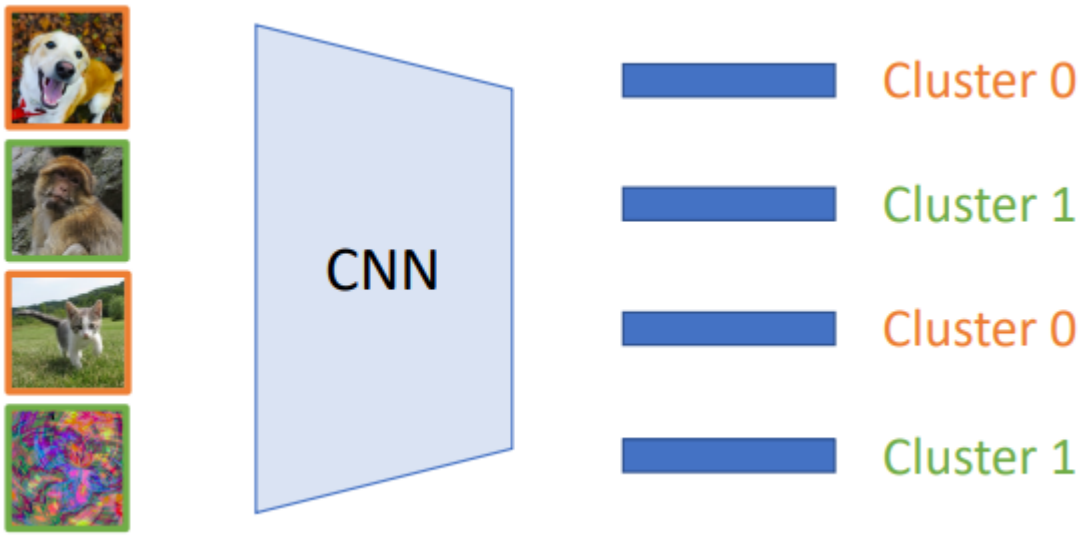

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(5) Repeat: GOTO (2)

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RotNet: Rotation Prediction

4-way classification task: How much was each image rotated CW? (0, 90, 180, or 270 degrees)

Gidaris et al, “Unsupervised representation learning by predicting image rotations”, ICLR 2018

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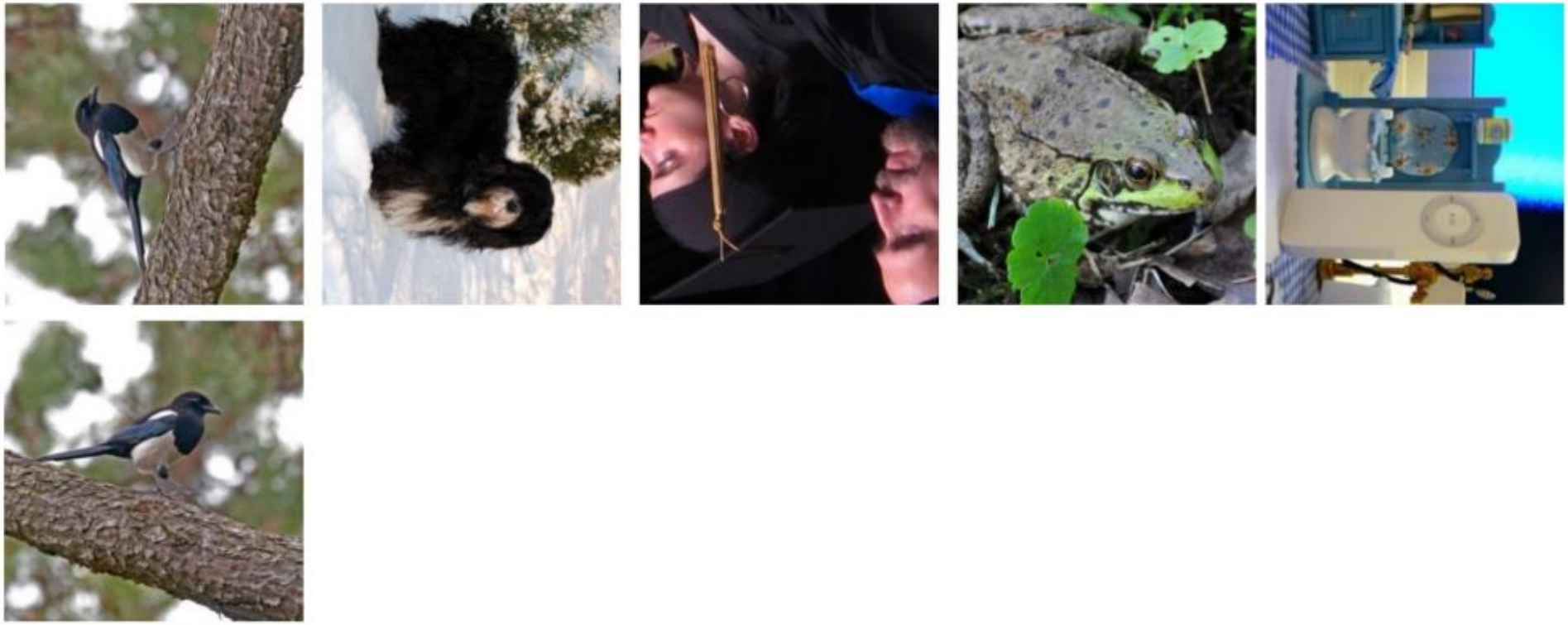

90

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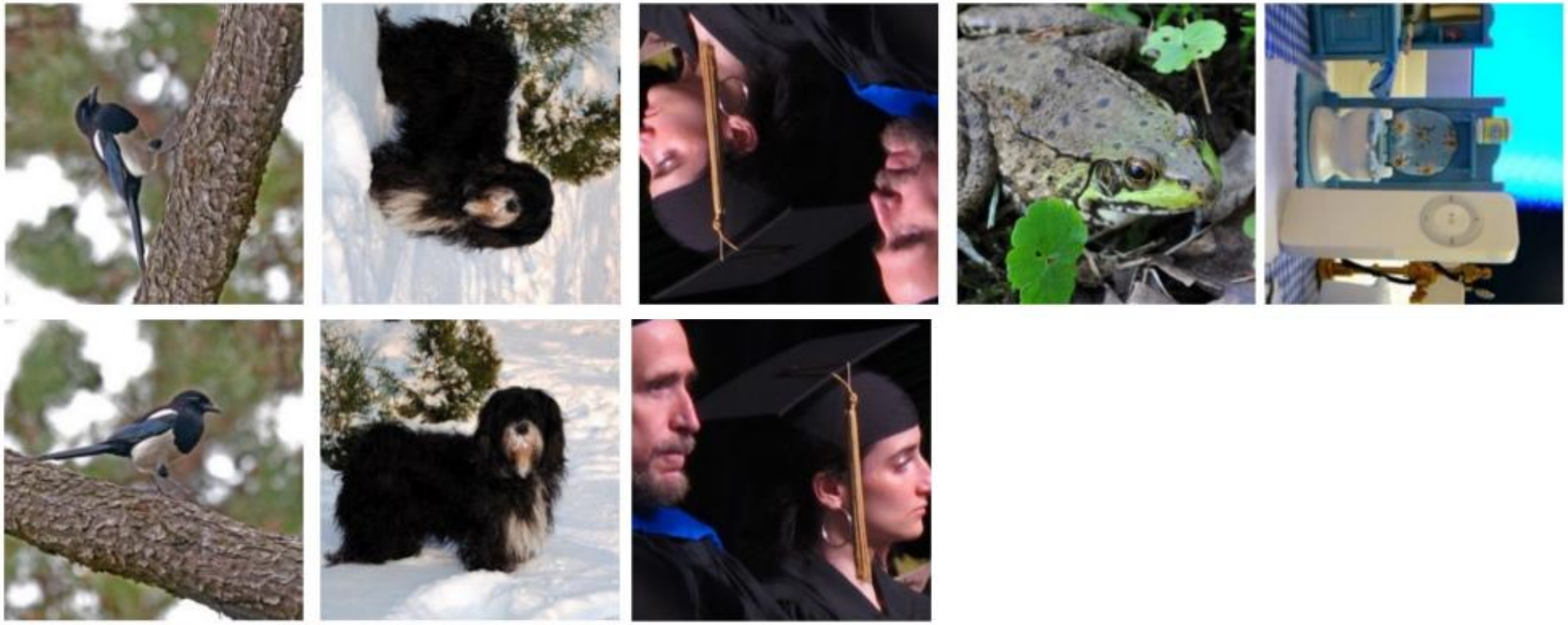

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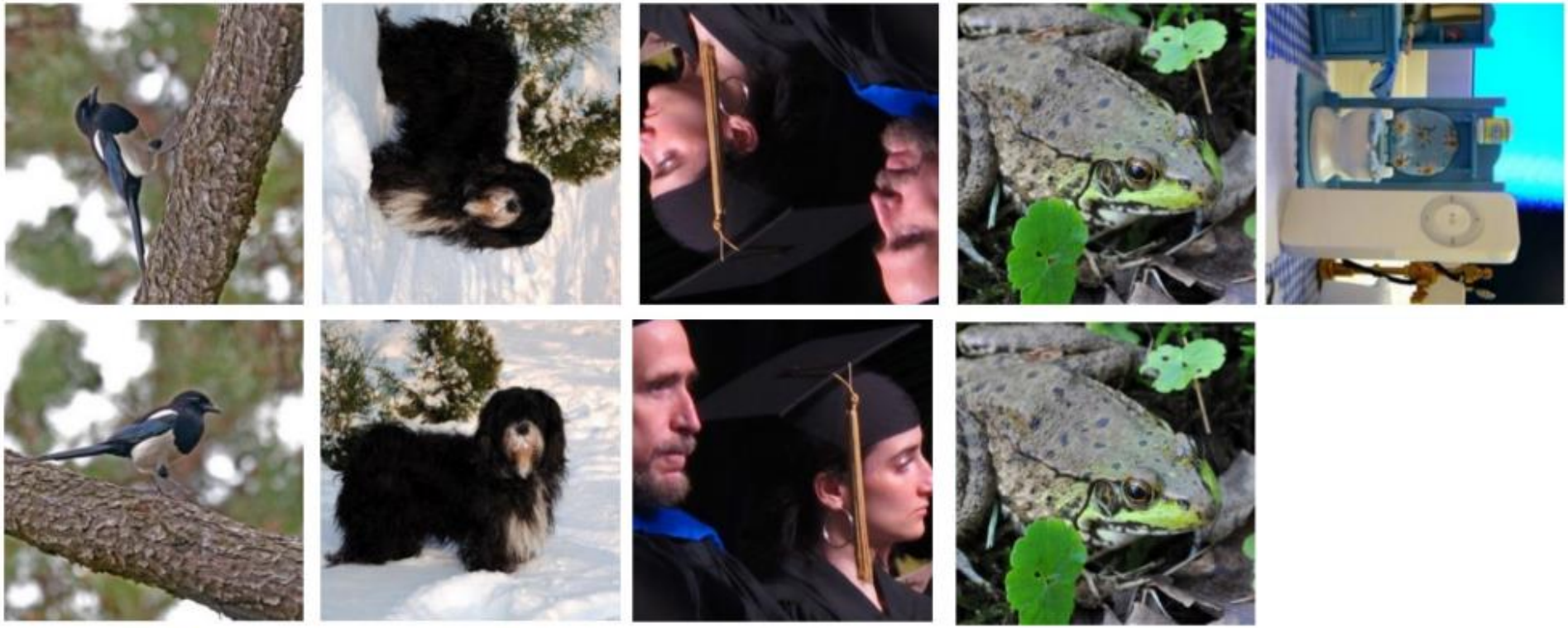

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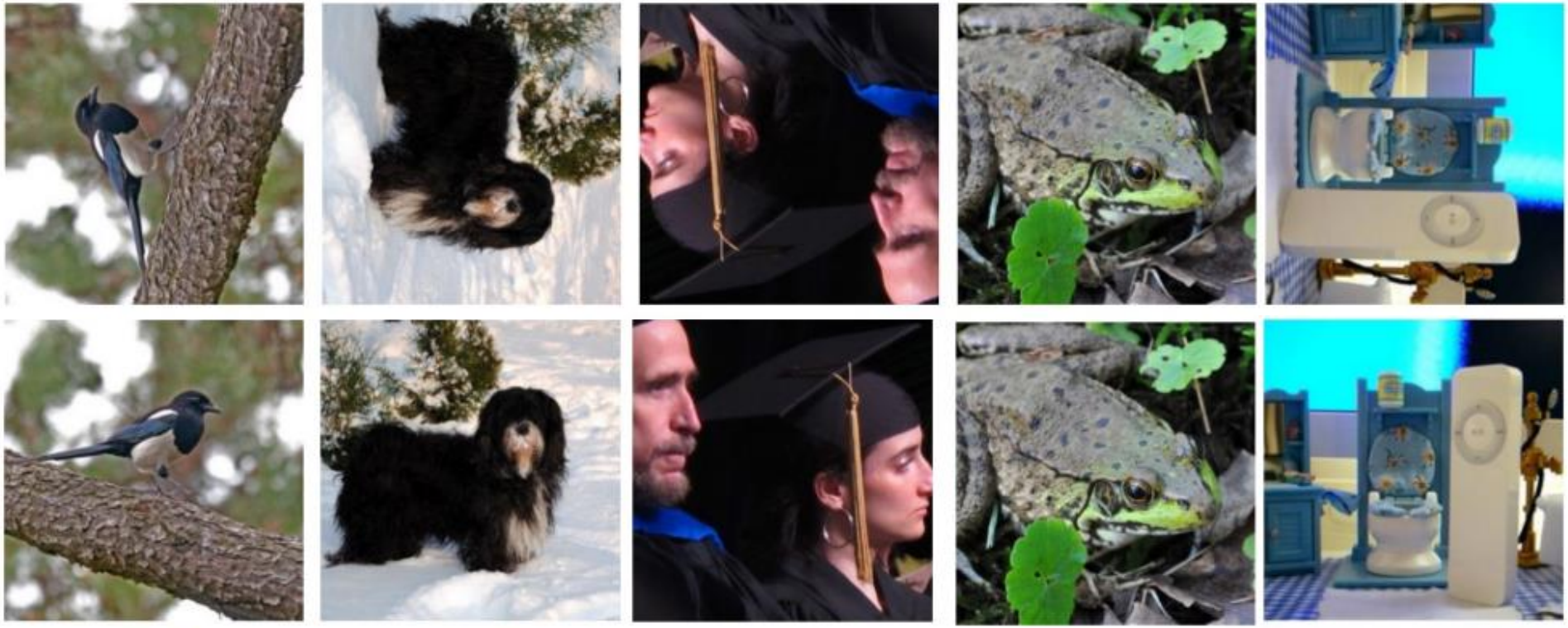

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Which SSL Method is Best?

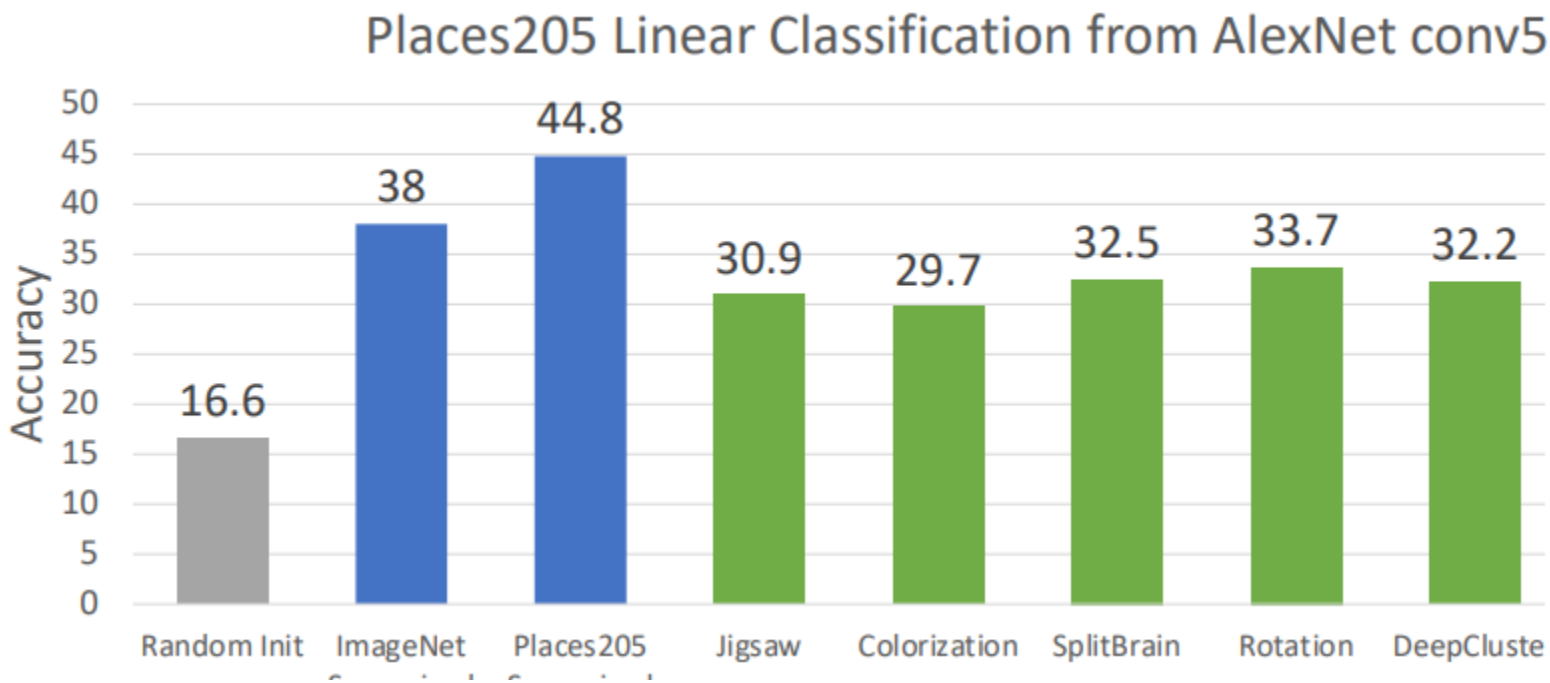
Fair evaluation of SSL methods is very hard! No theory , so we need to rely on experiment

Many choices in experimental setup, huge variations from paper to paper:

- CNN architecture? AlexNet, ResNet50, something else?
- Pretraining dataset? ImageNet, or something else?
- Downstream task? ImageNet classification, detection, something else?
- Pretraining hyperparameters? Learning rates, training iterations, data augmentation?
- Transfer learning protocol?
- - Linear probe? From which layer? How to train linear models? SGD, something else? Transfer learning hyperparameters? Data augmentation or BatchNorm during transfer learning?
- - Fine-tune? From which layer? Architecture of “head” you attach? Linear or nonlinear? Fine-tuning hyperparameters?
- - KNN? What value of K? Normalization on features?

Which SSL Method is Best?

Some papers have tried to do fair comparisons of many SSL methods



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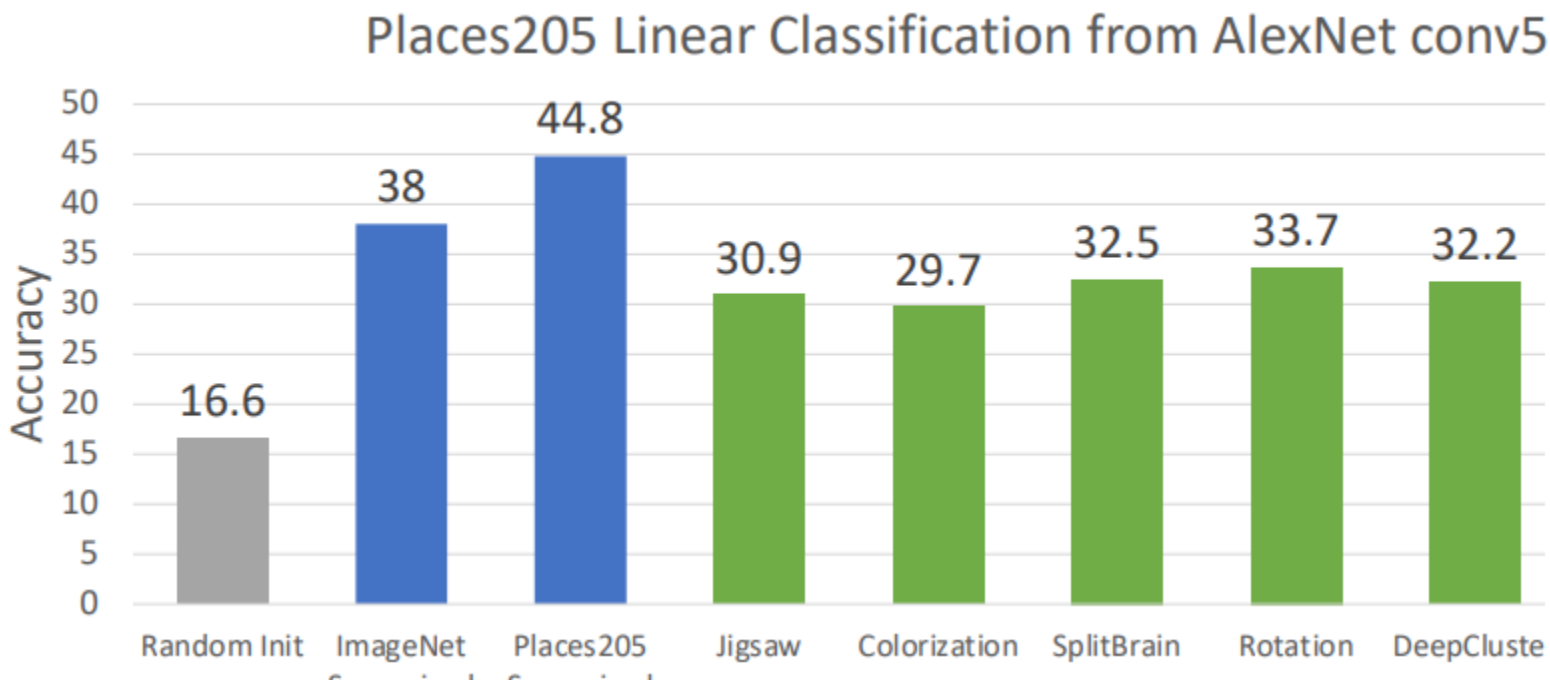
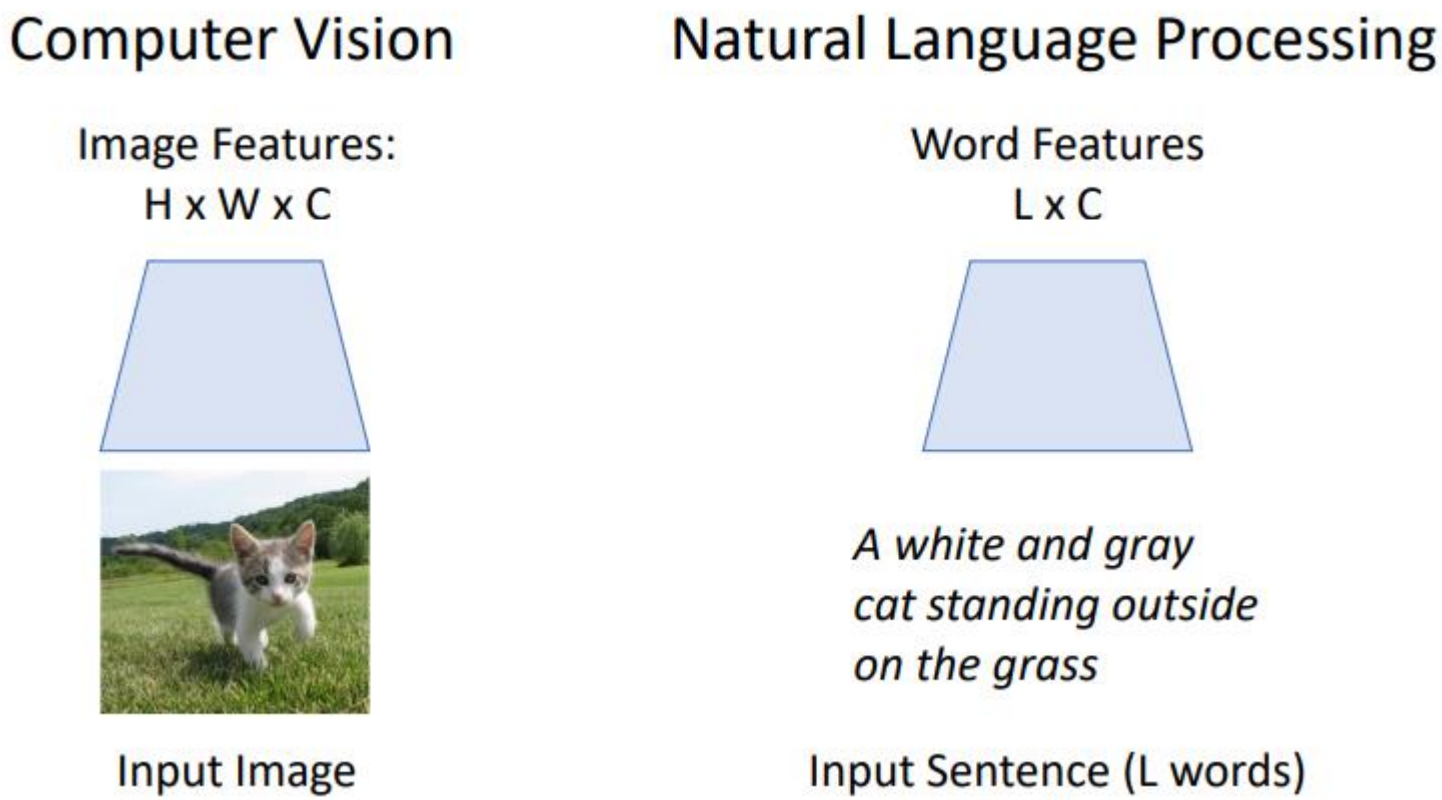


Overall, as of 2019, SSL gave worse features than supervised pretraining

Goyal et al, "Scaling and Benchmarking Self-Supervised Visual Representation Learning", ICCV 2019

Self-Supervised Learning for Natural Language



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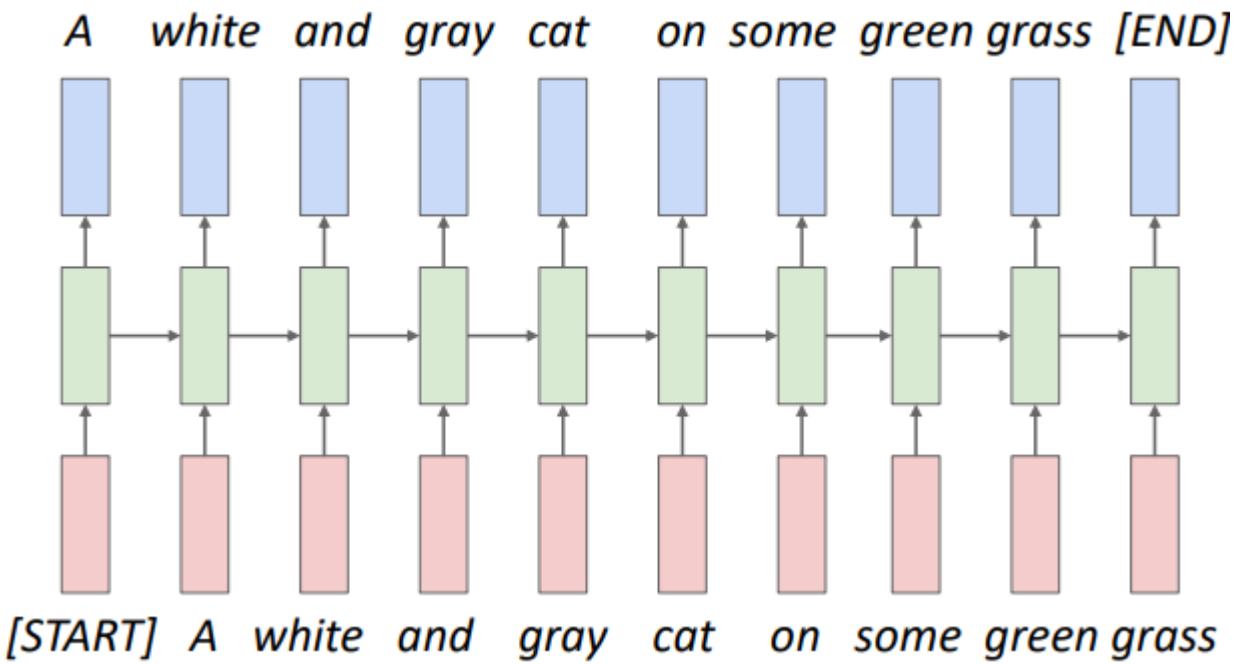
Self-Supervised Learning for Natural Language

RNN language models are trained on raw text – no human labels required!

Their hidden states give features that transfer to many downstream tasks!

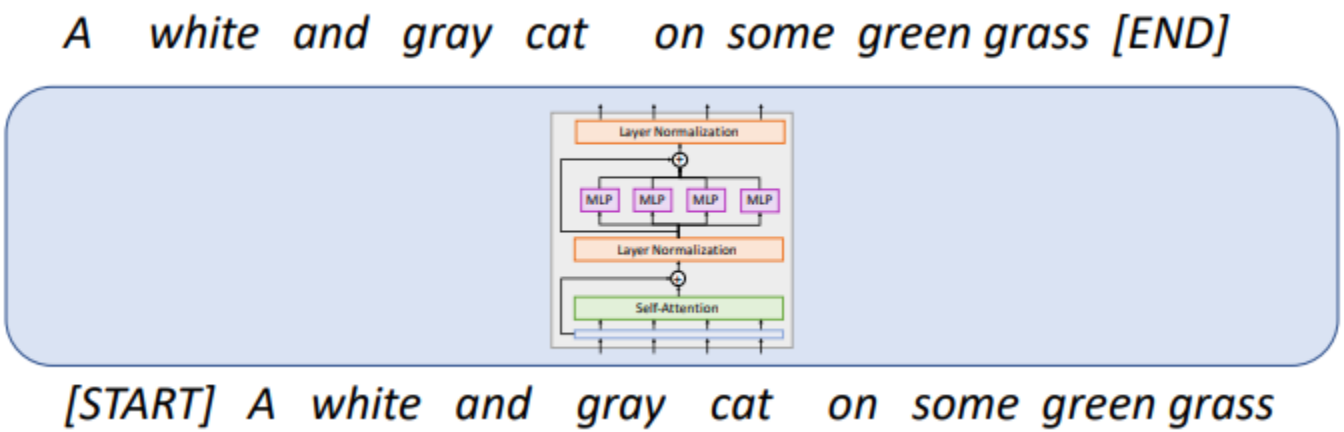


Peters et al, “Deep contextualized word representations”, NAACL 2018

Self-Supervised Learning for Natural Language

Transformer-based language models work even better! Can scale up to very large datasets, and give extremely powerful features that transfer to downstream tasks

Wildly successful: larger models, larger datasets give better features that improve performance on many downstream NLP tasks. The dream of SSL made real!



Radford et al, "Language models are unsupervised multitask learners", 2019
Brown et al, "Language Models are Few-Shot Learners", arXiv 2020
Rae et al, “Scaling Language Models: Methods, Analysis, & Insights from Training Gopher”, arXiv 2021

This Week: Pathways Language Model (PaLM)

Transformer with 118 layers, 48 heads, 540B parameters

Dataset: 780 billion tokens; trained on 6144 TPU-v4 chips

Bigger models trained on more data tend to give better downstream task performance

Model	Avg NLG	Avg NLU
GPT-3 175B	52.9	65.4
GLaM 64B/64E	58.4	68.7
PaLM 8B	41.5	59.2
PaLM 62B	57.7	67.3
PaLM 540B	63.9	74.7

NLG = Natural Language Generation (8 benchmarks)

NLU = Natural Language Understanding (21 benchmarks)

Chowhery et al, "PaLM: Scaling Language Models with Pathways", 2022

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How can we achieve this success in vision?
Intensified interest in SSL since ~2018

Chowhery et al, "PaLM: Scaling Language Models with Pathways", 2022

Exemplar CNN: Invariance to Data Augmentation

Quiz: What is this?

Dosovitskiy et al, “Discriminative Unsupervised Feature Learning with Exemplar Convolutional Neural Networks”

Exemplar CNN: Invariance to Data Augmentation

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Answer: Deer!

Dosovitskiy et al, “Discriminative Unsupervised Feature Learning with Exemplar Convolutional Neural Networks”

Exemplar CNN: Invariance to Data Augmentation

Quiz: What is this?

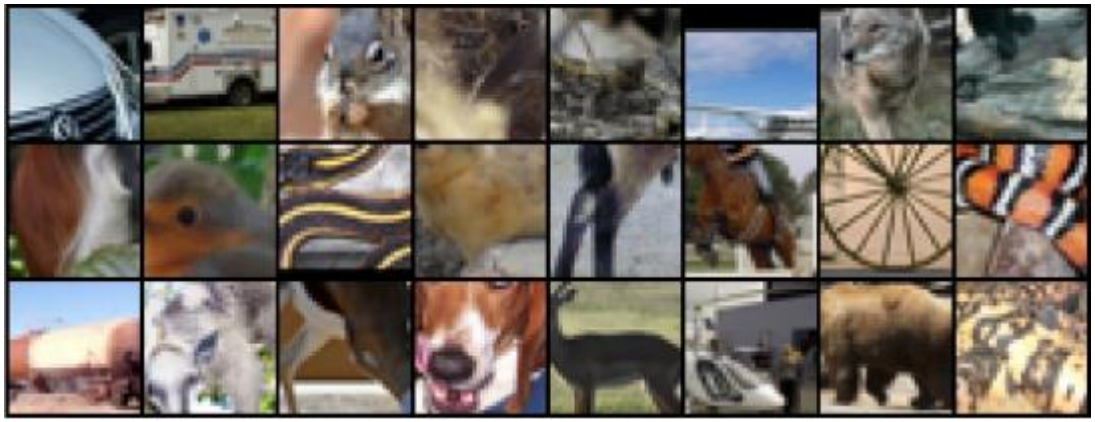
Answer: Deer!

Different data augmentations (scale, shift, color jitter) of the same initial image patch

Dosovitskiy et al, “Discriminative Unsupervised Feature Learning with Exemplar Convolutional Neural Networks”

Exemplar CNN: Invariance to Data Augmentation

Given an initial dataset of N image patches



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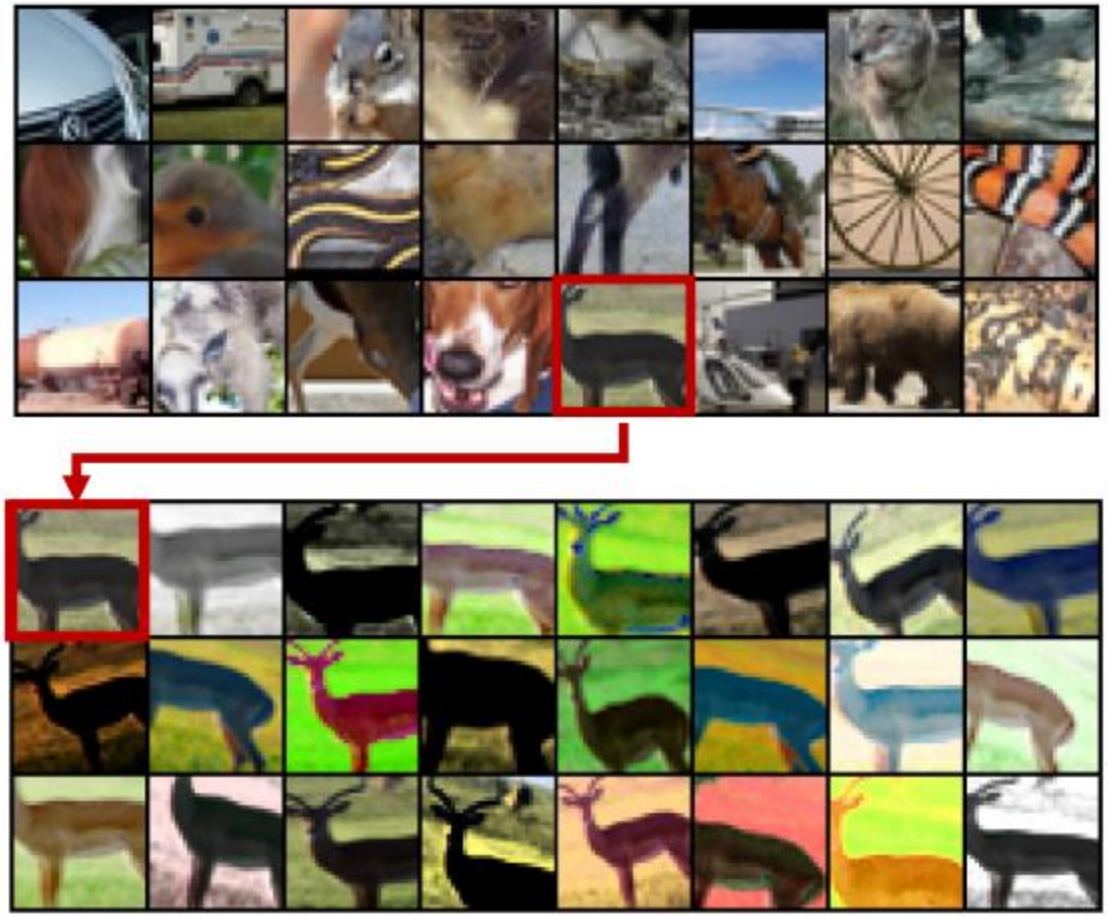

Sample K different augmentations for each; now have K*N total patches

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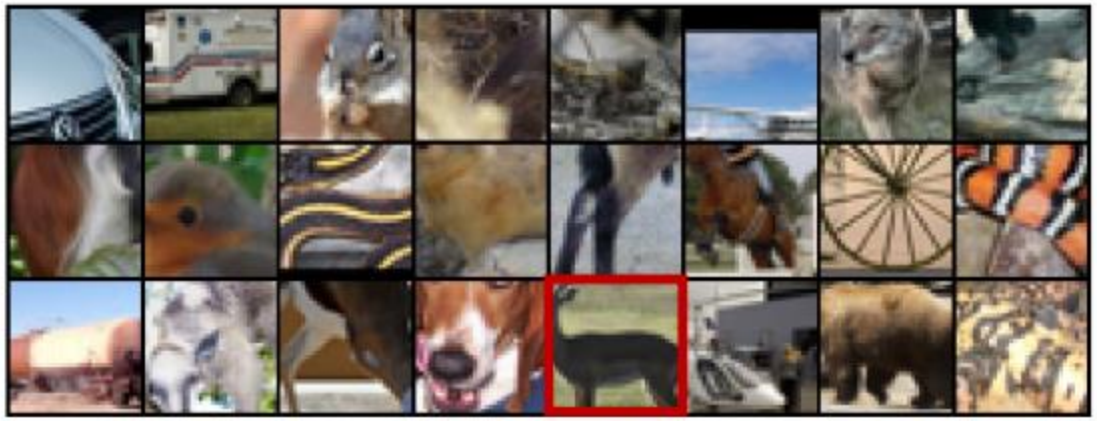

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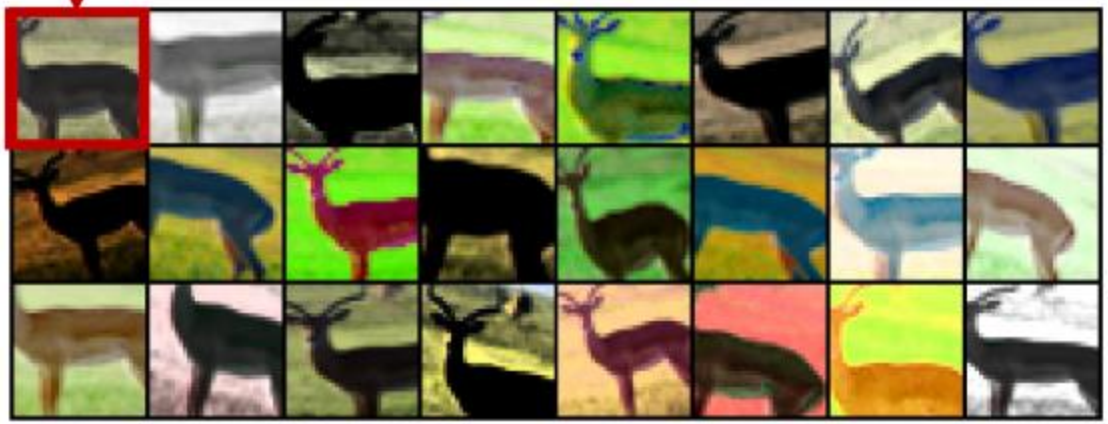

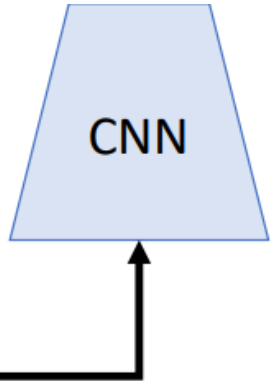

An augmented patch is fed into CNN

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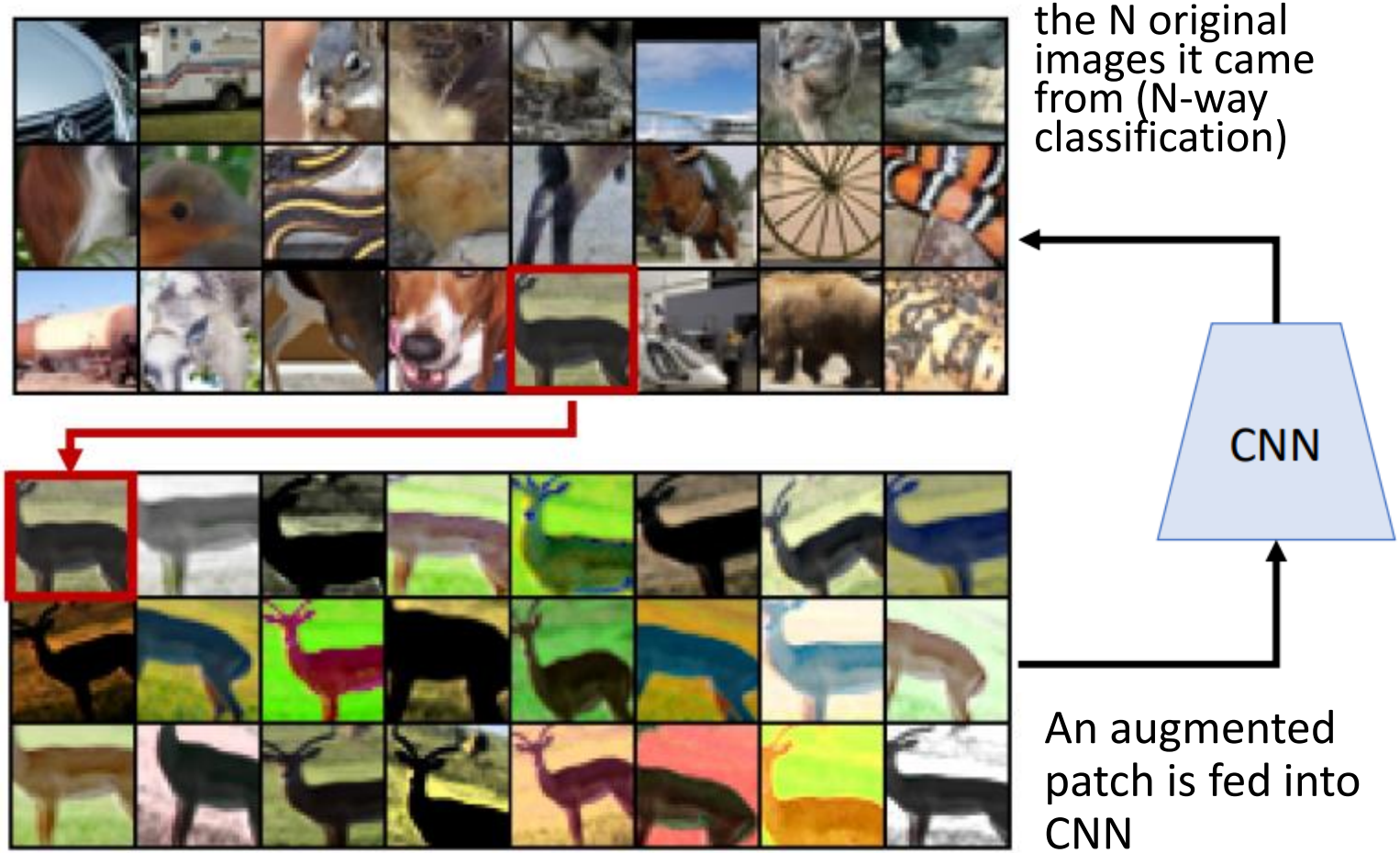
Exemplar CNN: Invariance to Data Augmentation

Given an initial dataset of N image patches

Predicts which of the N original images it came from (N-way classification)



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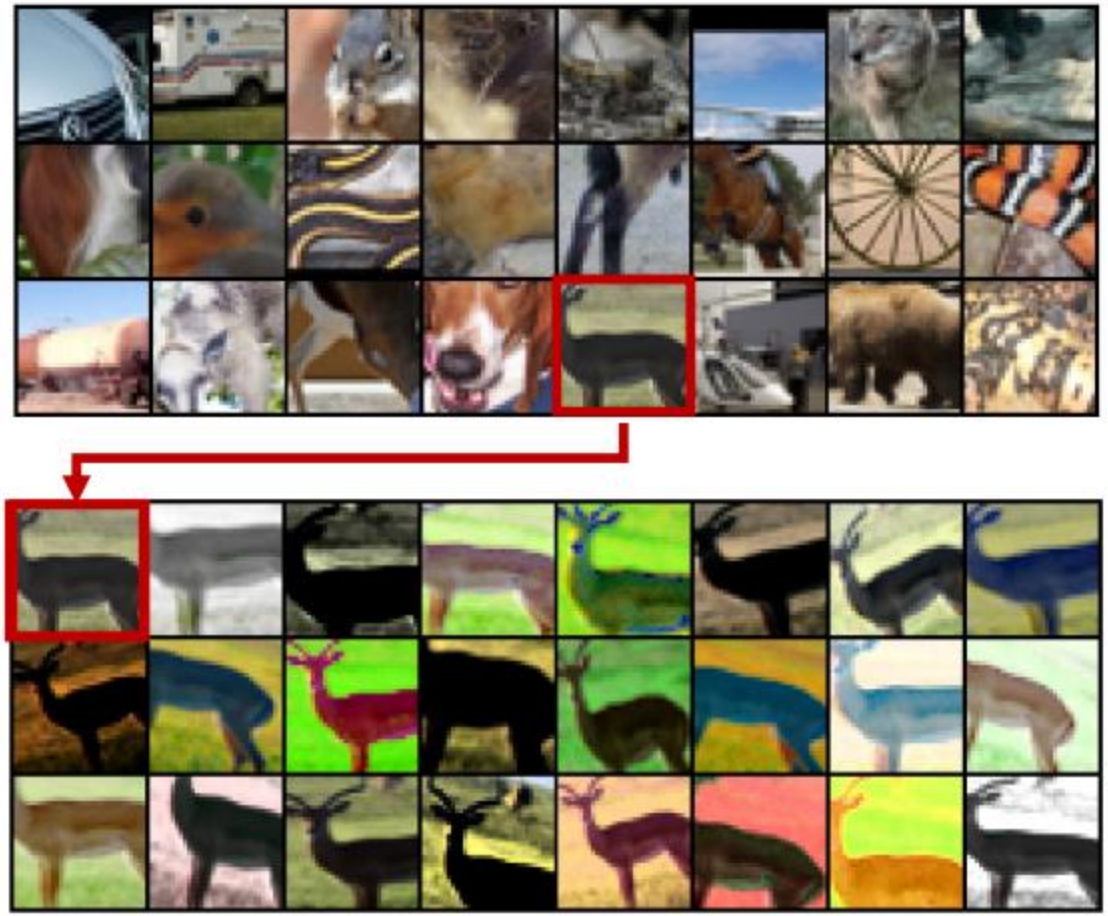
Exemplar CNN: Invariance to Data Augmentation

Problem: number of parameters in final layer depends on N; hard to scale

Given an initial dataset of N image patches



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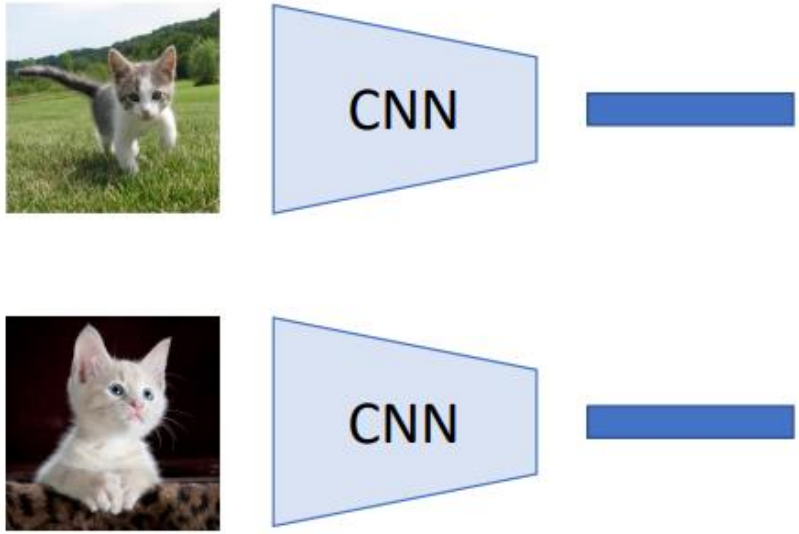
Contrastive Learning

Assume we don't have labels for images, but we know whether some pairs of images are **similar** or **dissimilar**

Similar images should have similar features



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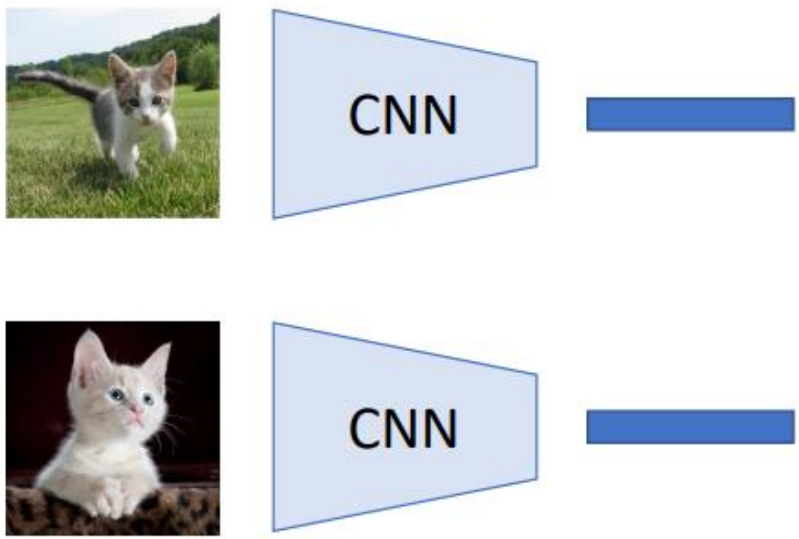
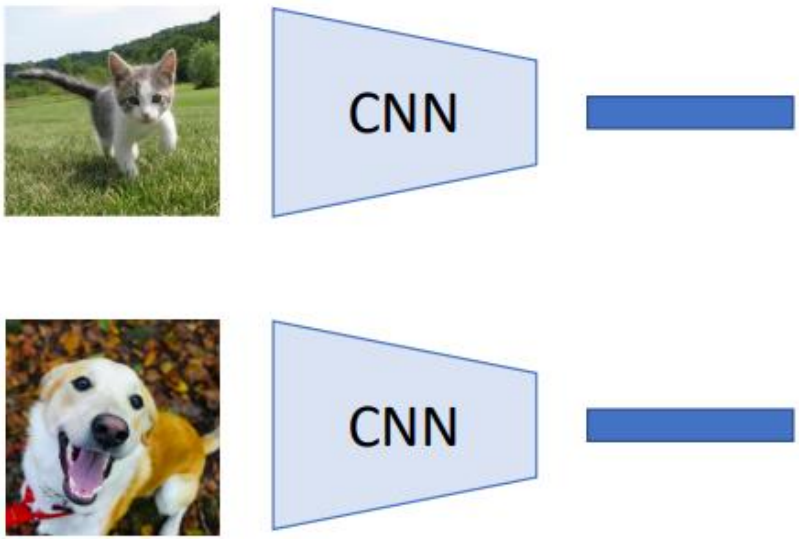


Dissimilar images should have dissimilar features



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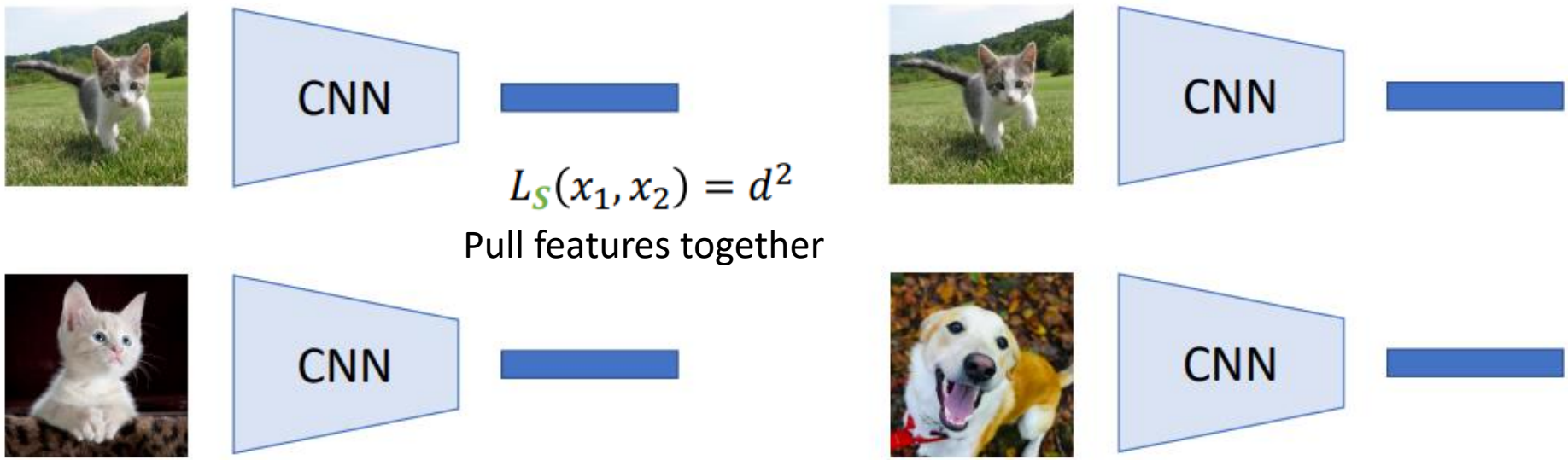
Contrastive Learning

Assume we don't have labels for images, but we know whether some pairs of images are **similar** or **dissimilar**

Let d= $\|\phi(x_1) - \phi(x_2)\|_2$ be the Euclidean distance between features for two images

Similar images should have similar features

Dissimilar images should have dissimilar features



$$L_S(x_1, x_2) = d^2$$

Pull features together

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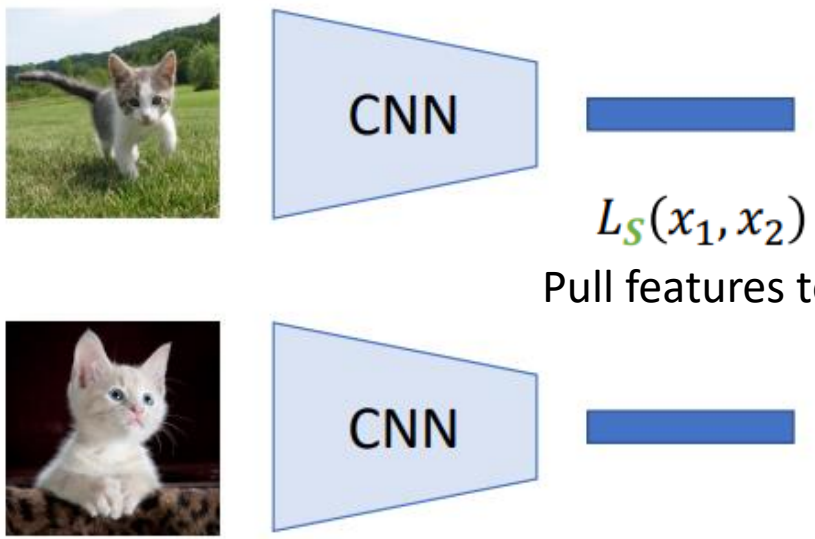

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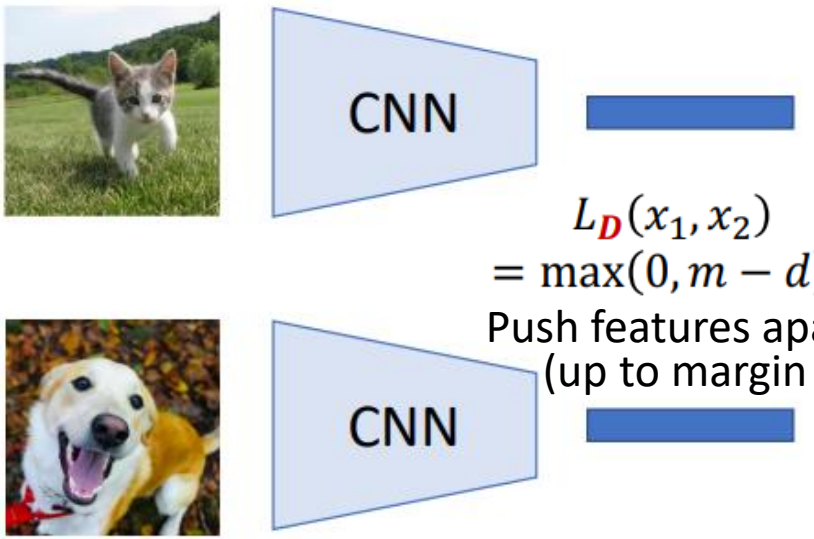


$L_D(x_1, x_2)$
$= \max(0, m - d)^2$

Push features apart (up to margin m)

Hadsell et al, "Dimensionality Reduction by Learning and Invariant Mapping", CVPR 2006

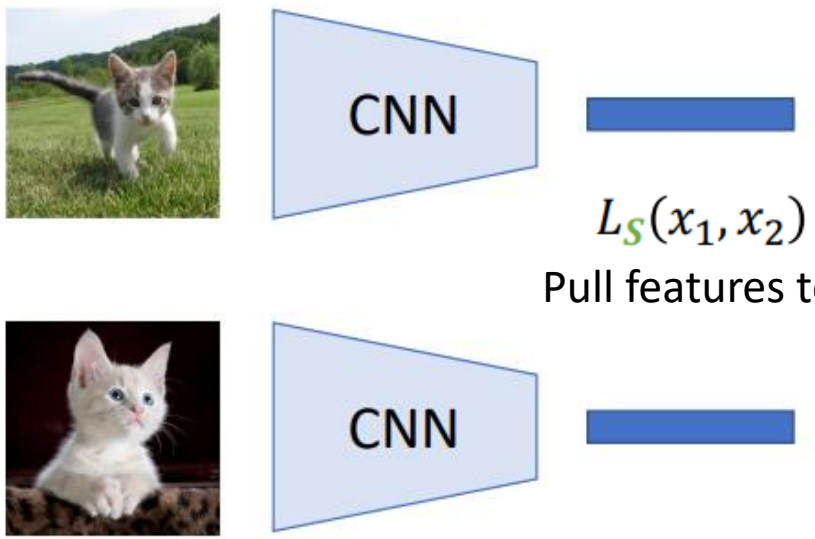
Contrastive Learning

Assume we don't have labels for images, but we know whether some pairs of images are **similar** or **dissimilar**

Problem: Where to get positive and negative pairs?

Let d= $\|\phi(x_1) - \phi(x_2)\|_2$ be the Euclidean distance between features for two images

Similar images should have similar features



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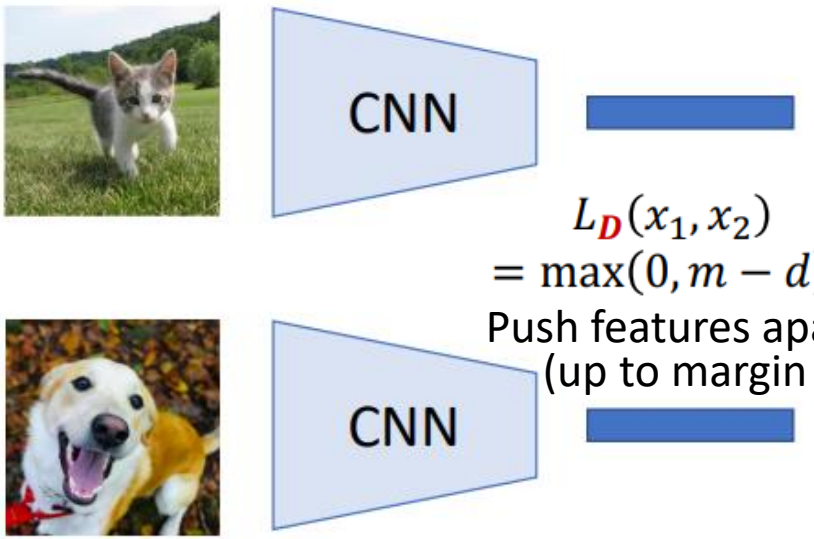

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Contrastive Learning with Data Augmentation

Batch of
N images

Hadsell et al, “Dimensionality Reduction by Learning and Invariant Mapping”, CVPR 2006
Wu et al, “Unsupervised Feature Learning by Non-Parametric Instance-Level Discrimination”, CVPR 2018
Van den Oord et al, “Representation Learning with Contrastive Predictive Coding”, NeurIPS 2018

Hjelm et al, “Learning deep representations by mutual information estimation and maximization”, ICLR 2019 Bachman et al, “Learning Representations by Maximizing Mutual Information Across Views”, NeurIPS 2019 Henaff et al, “Data-Efficient Image Recognition with Contrastive Predictive Coding”, ICML 2020

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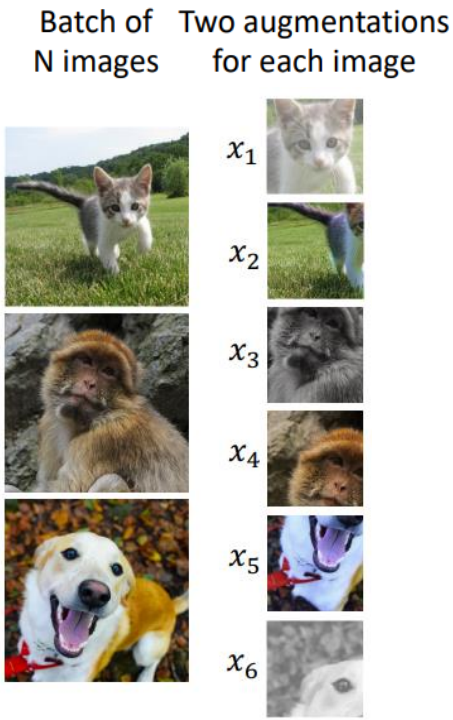

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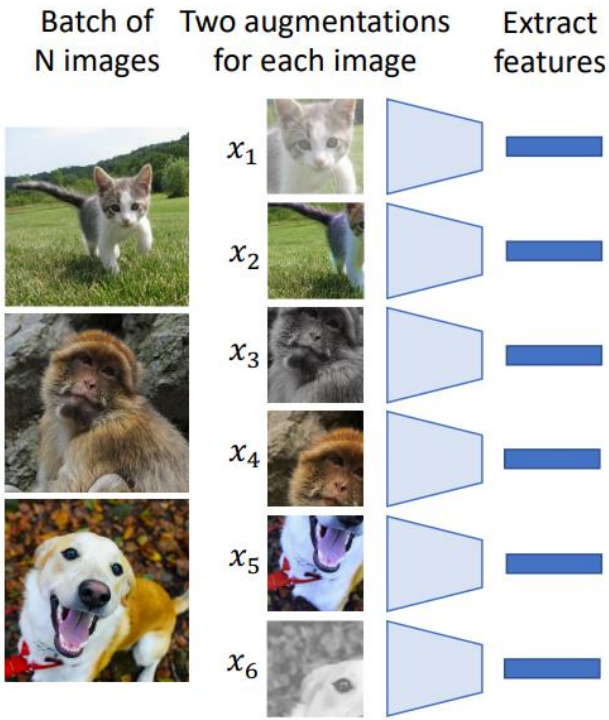

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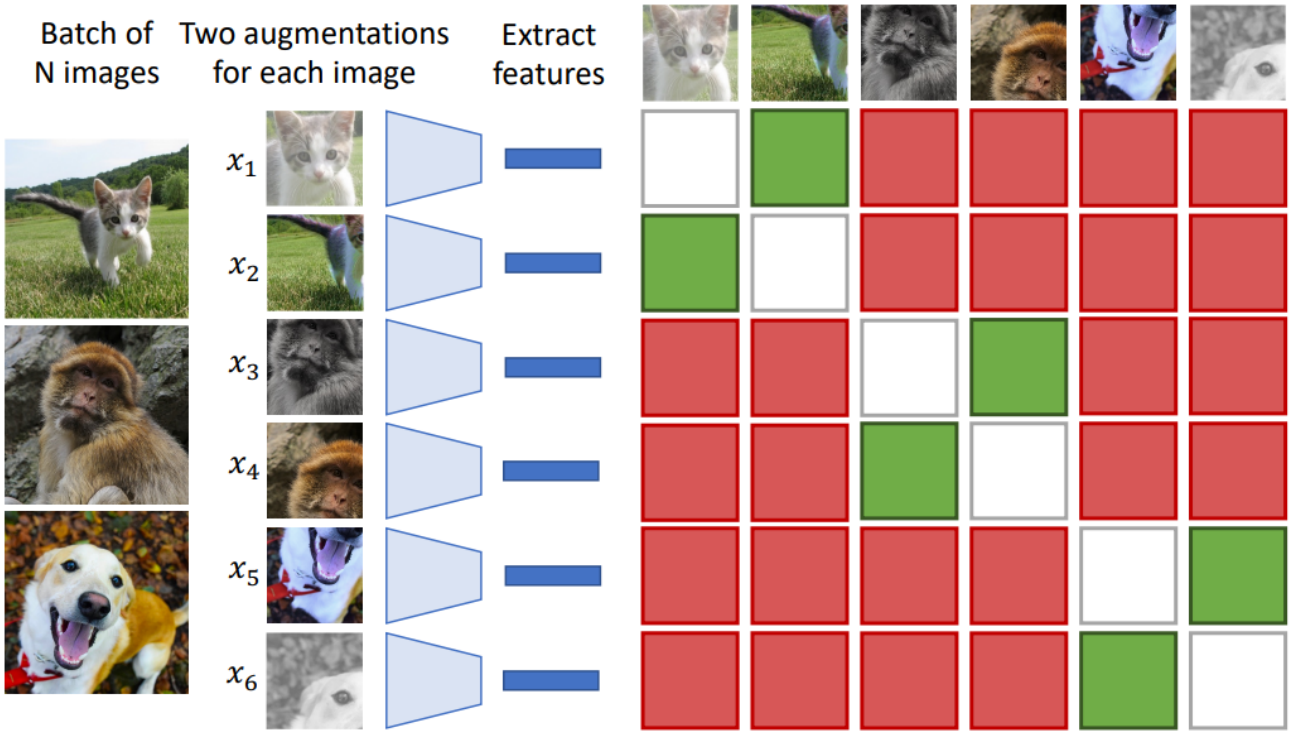
Contrastive Learning with Data Augmentation



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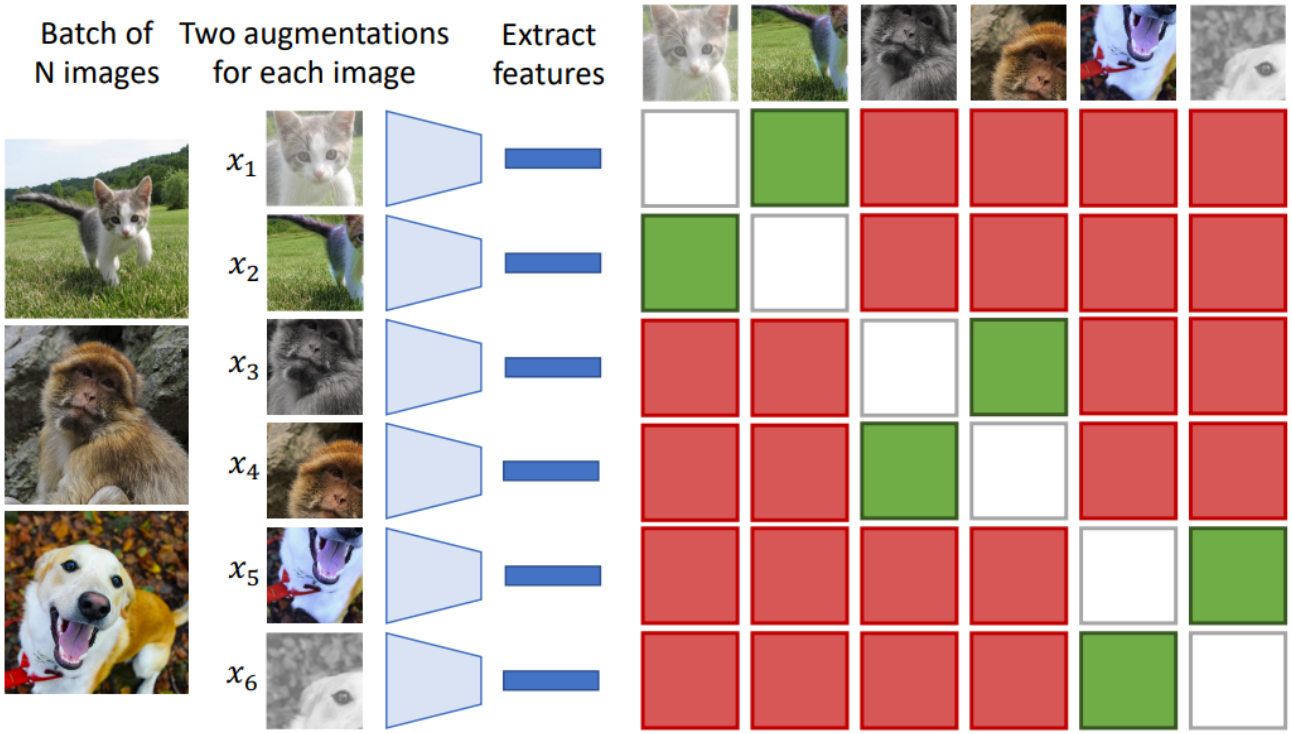

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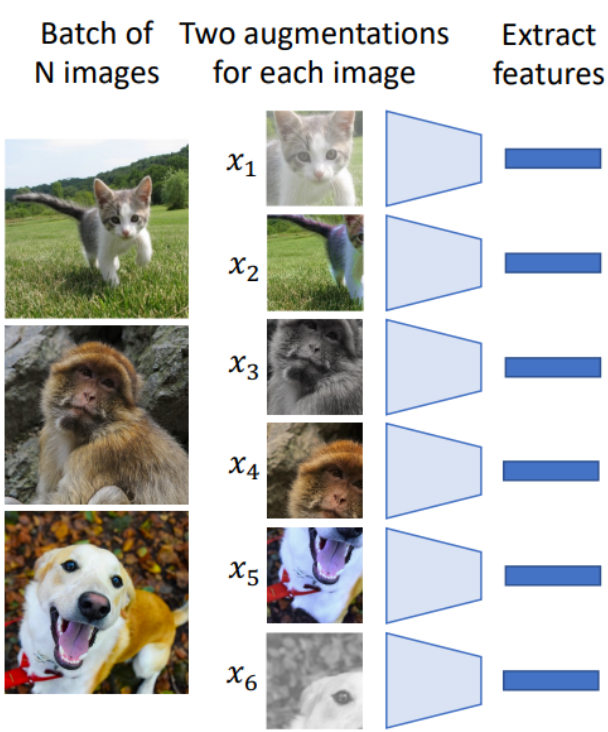

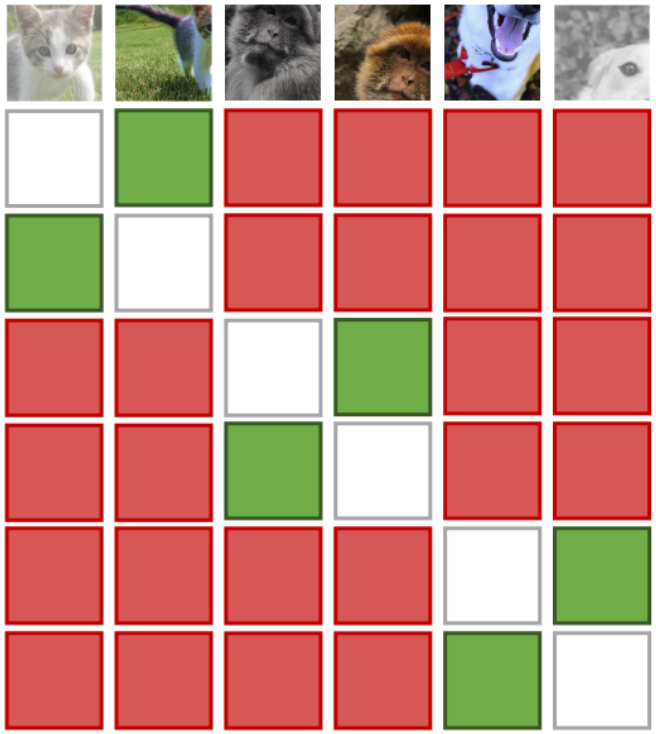

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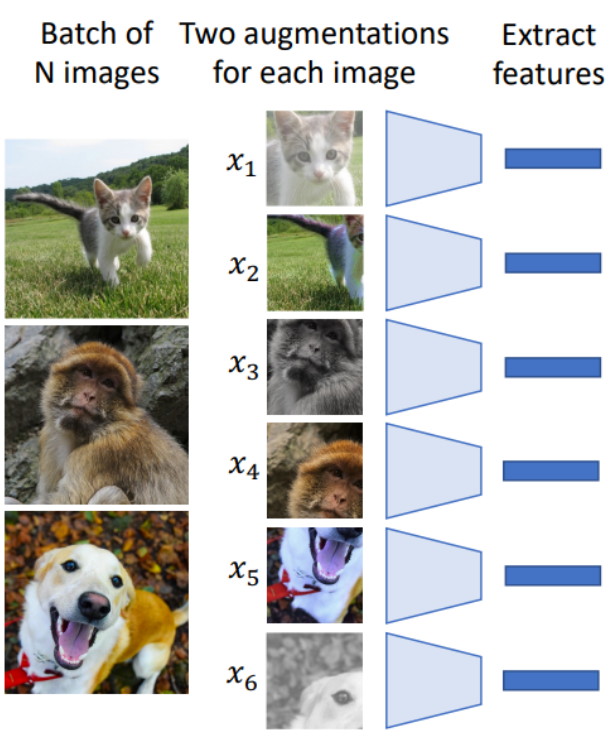

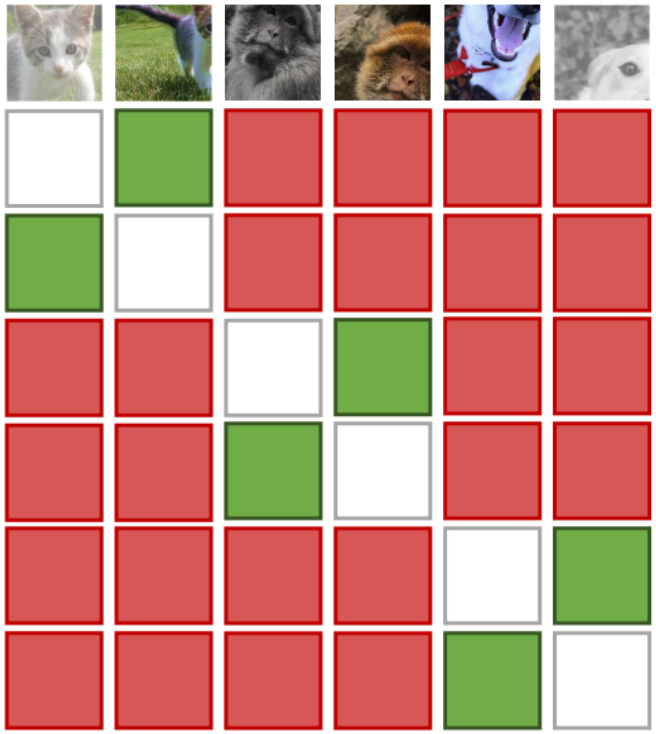

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Interpretation: Cross-entropy loss over the other 2N-1 elements in the batch!

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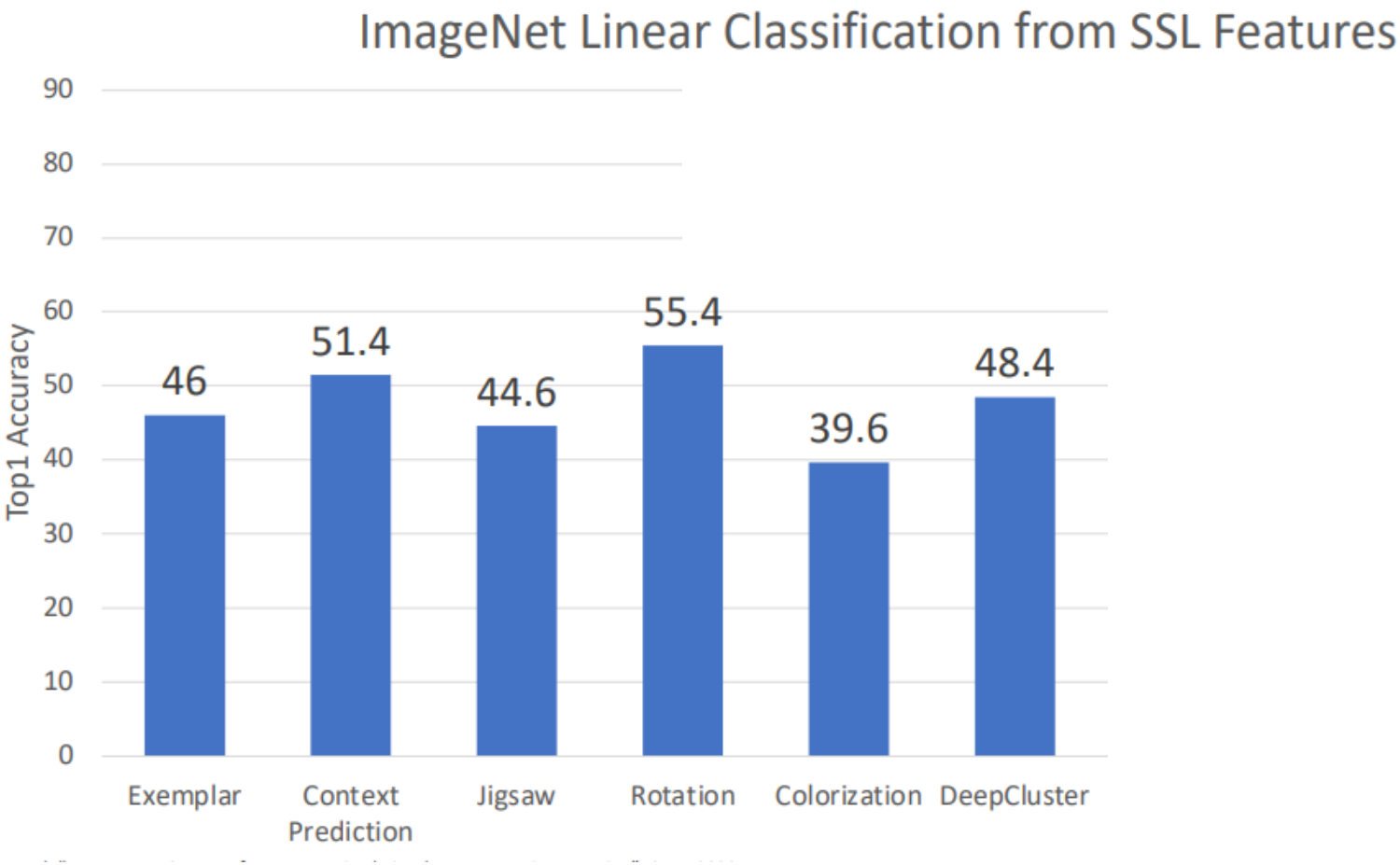

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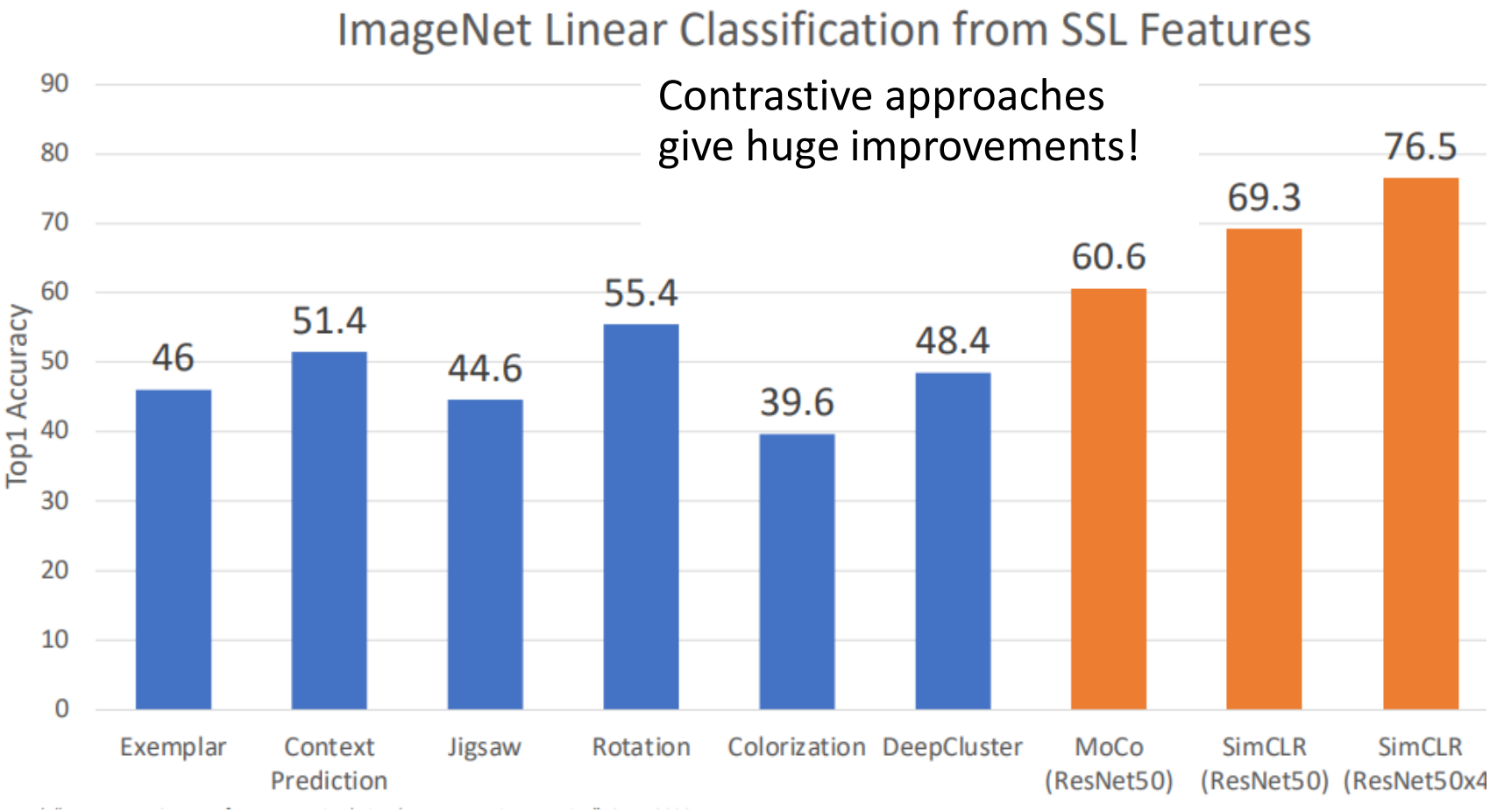
(Lots of caveats here, such as different architectures, etc.)



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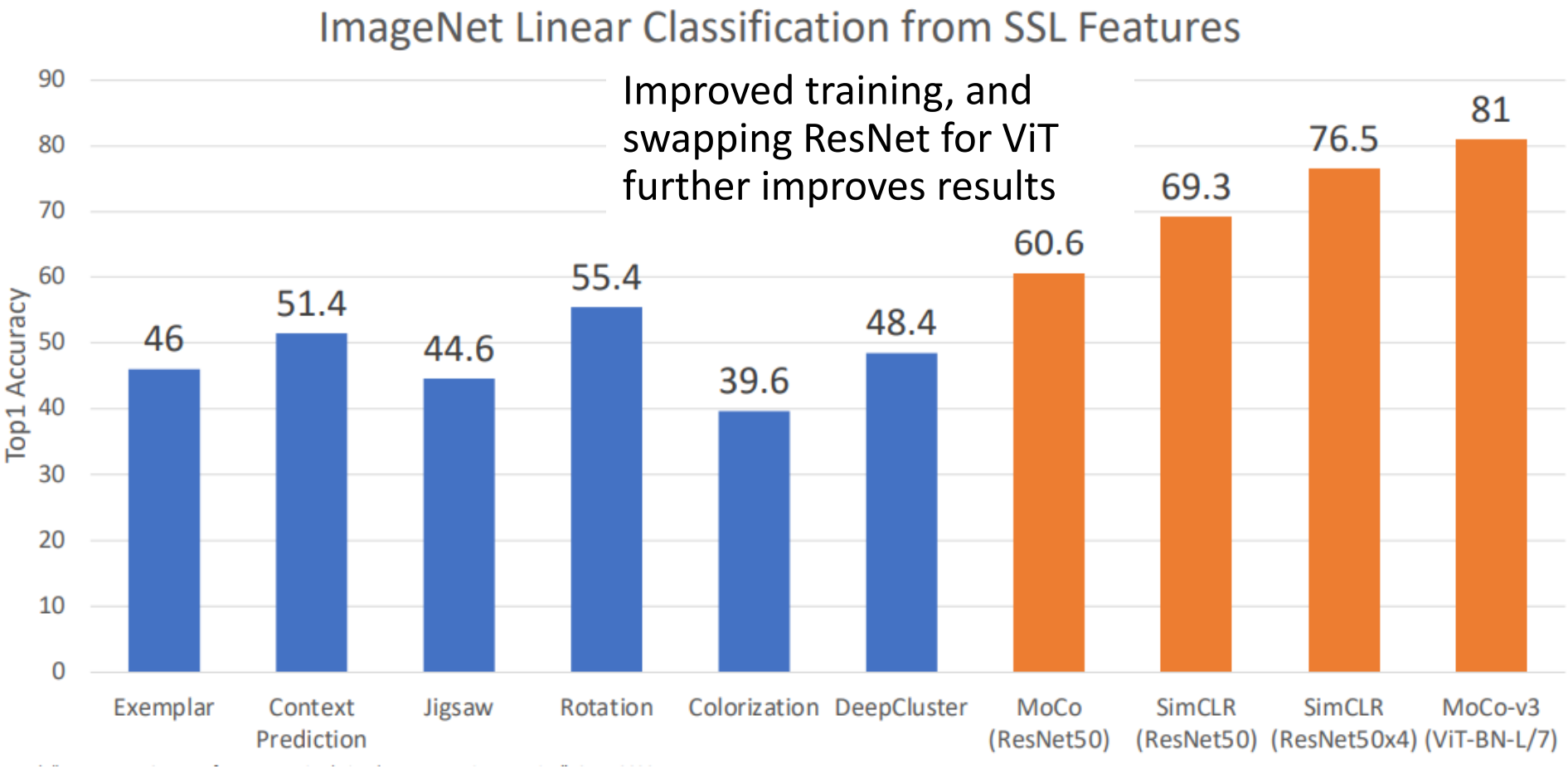

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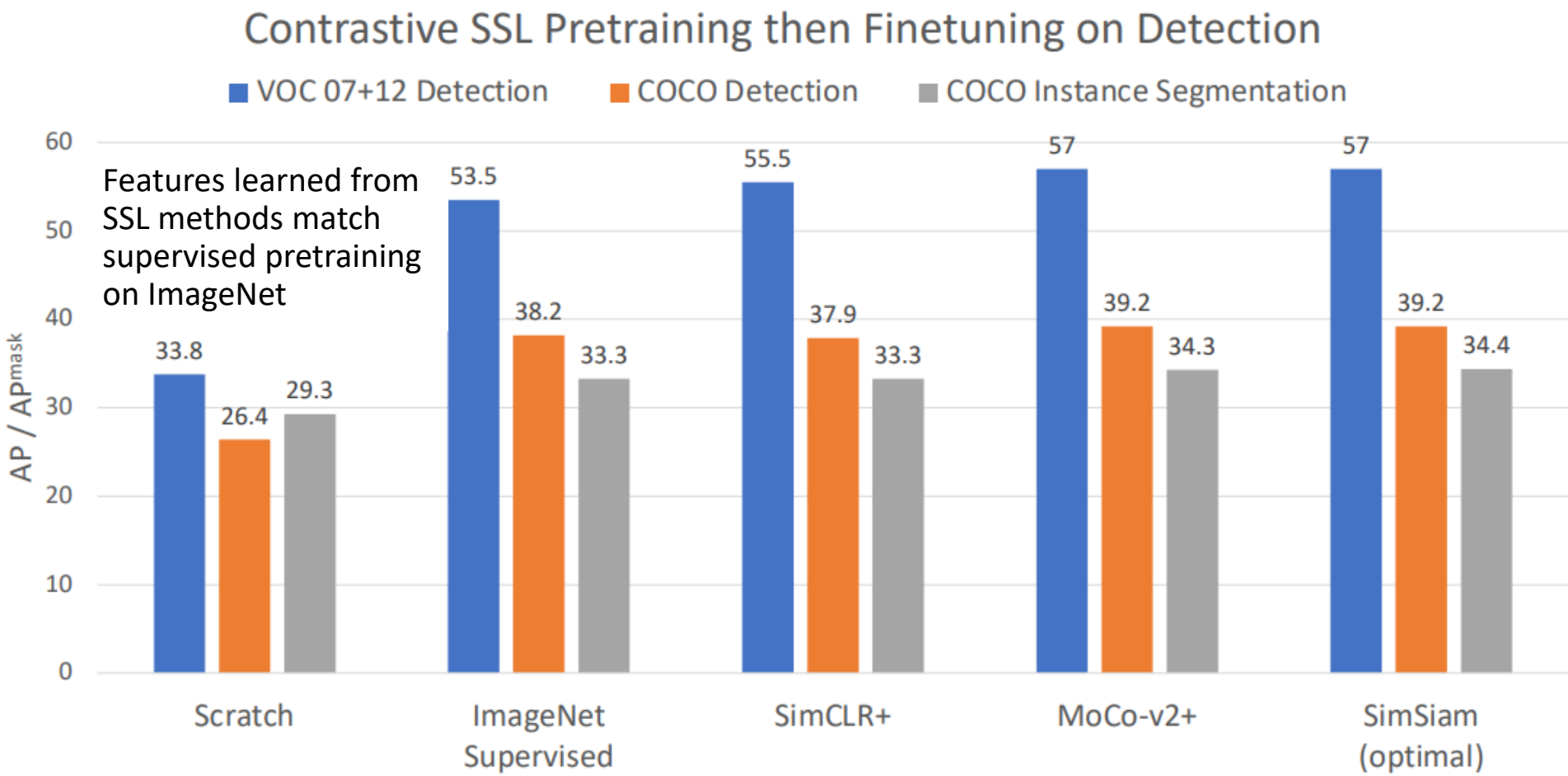

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Masked Autoencoders (MAE)

A ~~new~~ old method dethrones contrastive learning? Denoising Autoencoder with Vision Transformer

He et al, “Masked Autoencoders are Scalable Vision Learners”, CVPR 2022

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Divide image into nonoverlapping patches, discard most of them

input

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Encode remaining patches with a ViT

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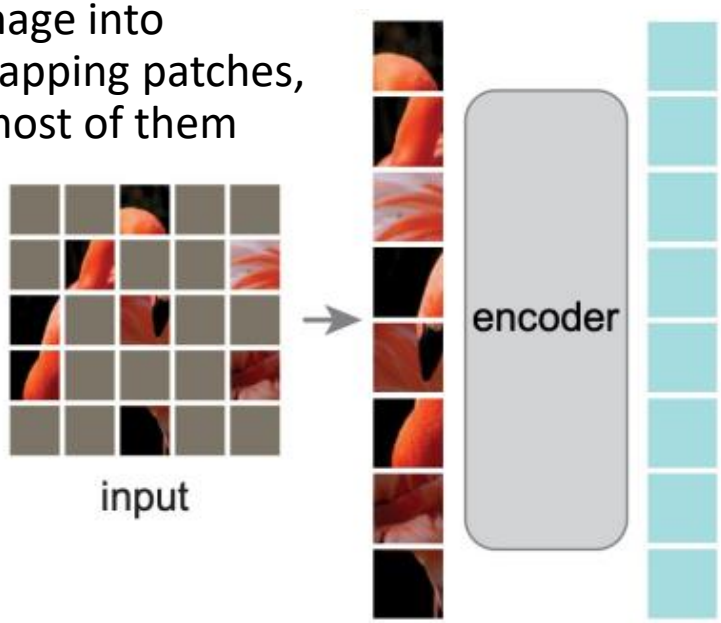

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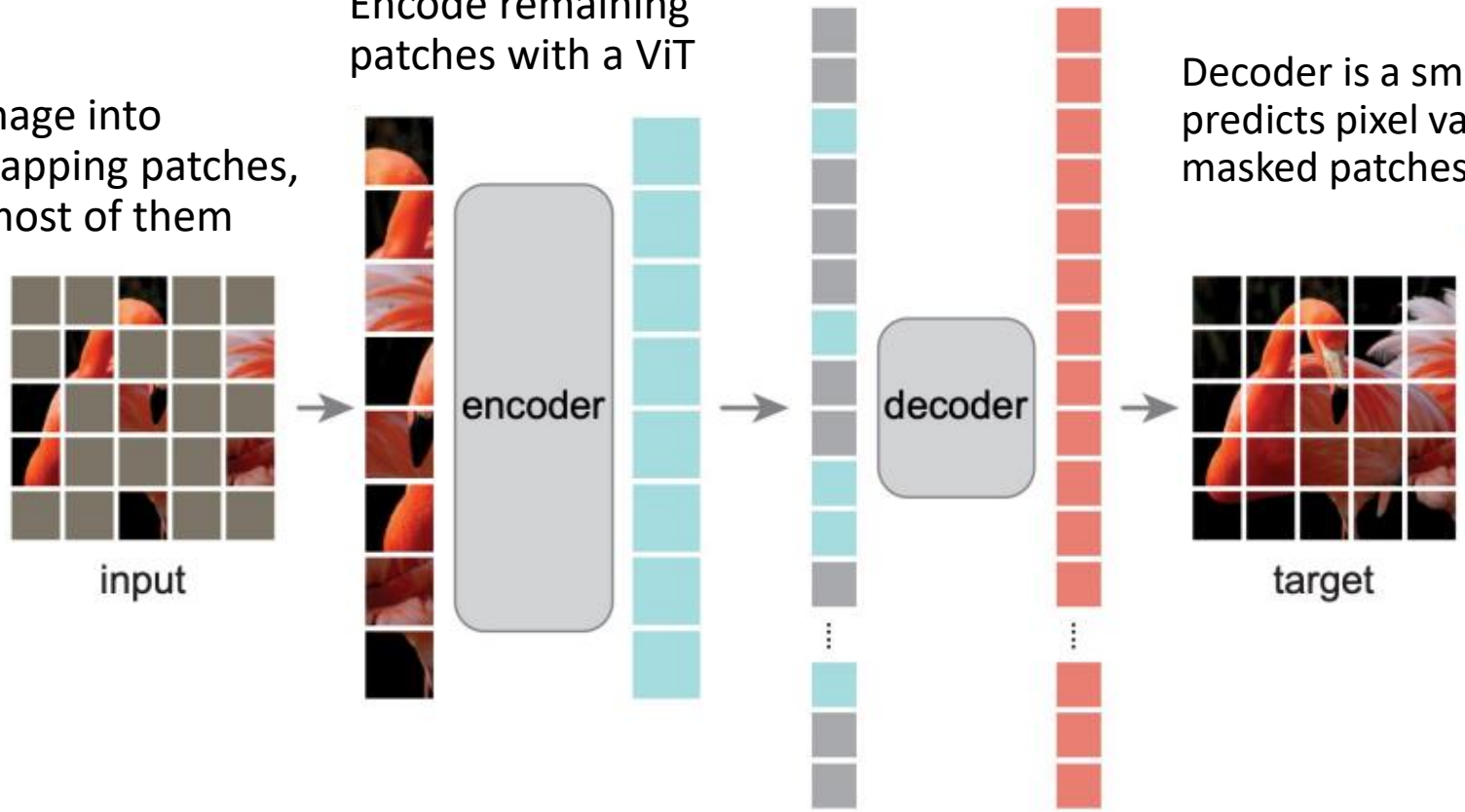
Encode remaining patches with a ViT

Decoder is a small ViT that predicts pixel values of the masked patches



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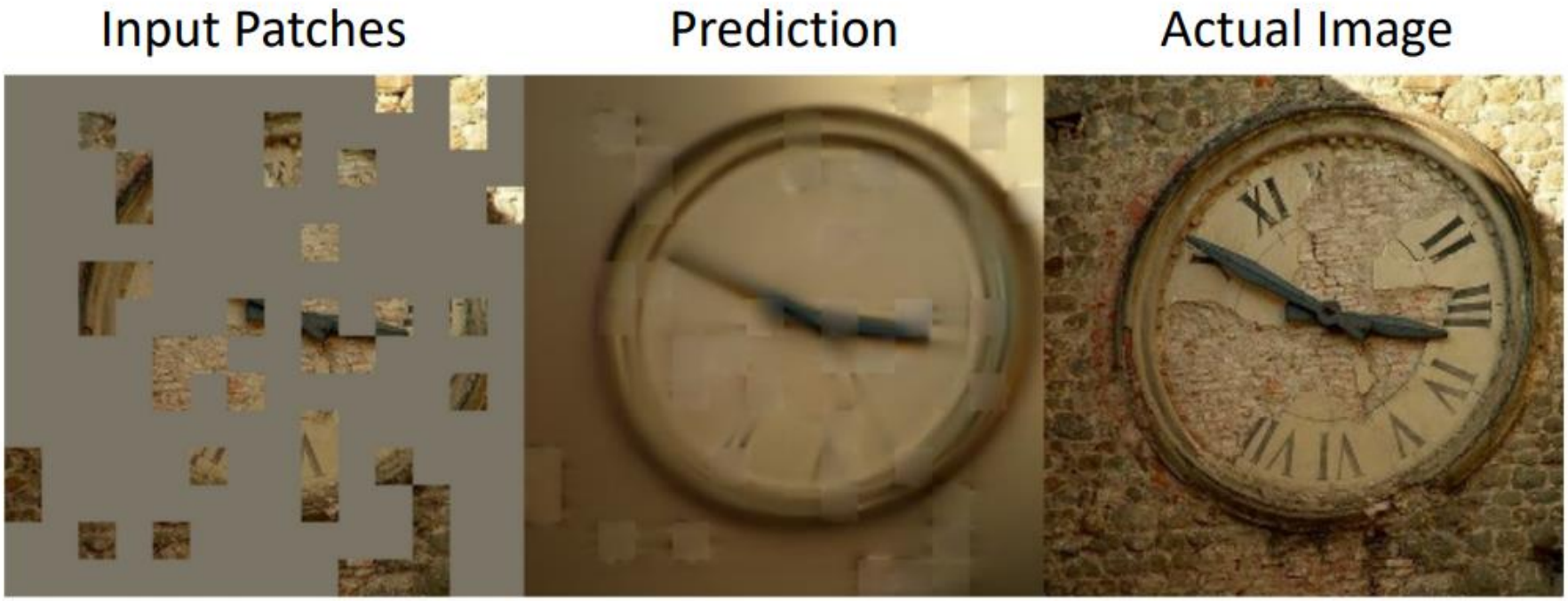
Masked Autoencoders (MAE): Reconstructions



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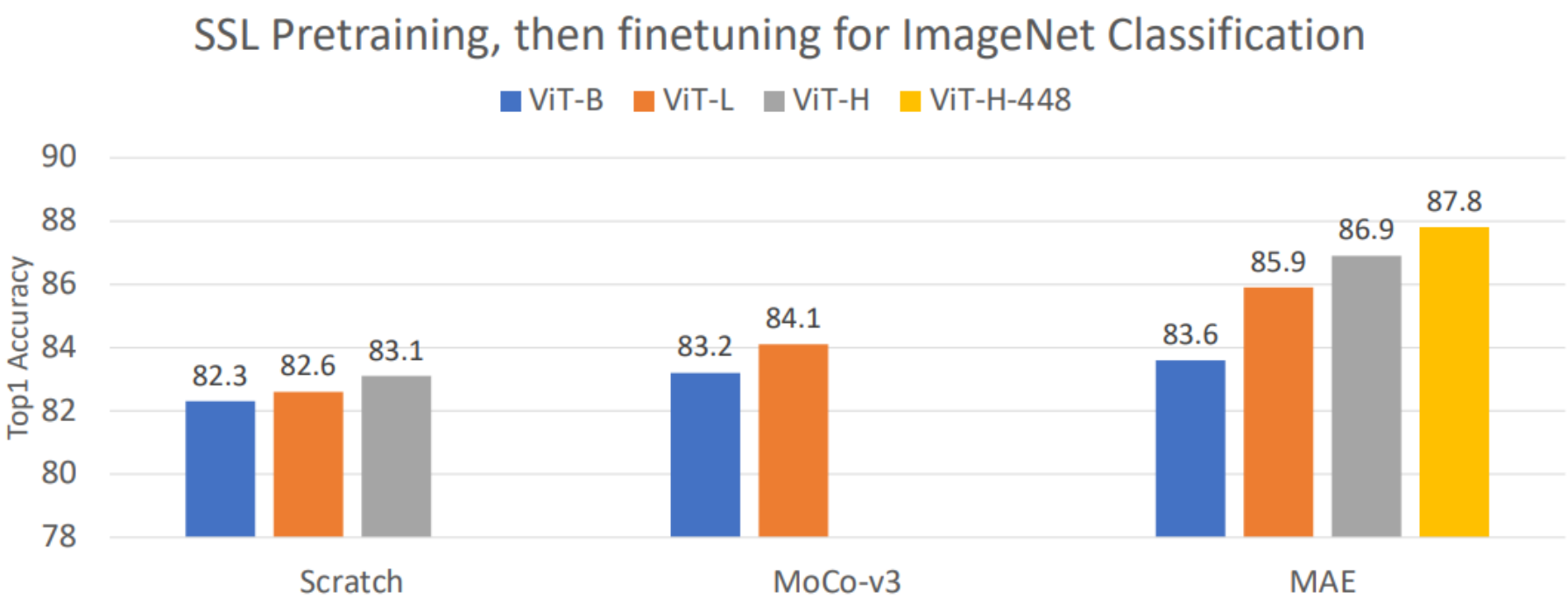

MAE pretraining outperforms training from scratch, and allows scaling to larger ViT models

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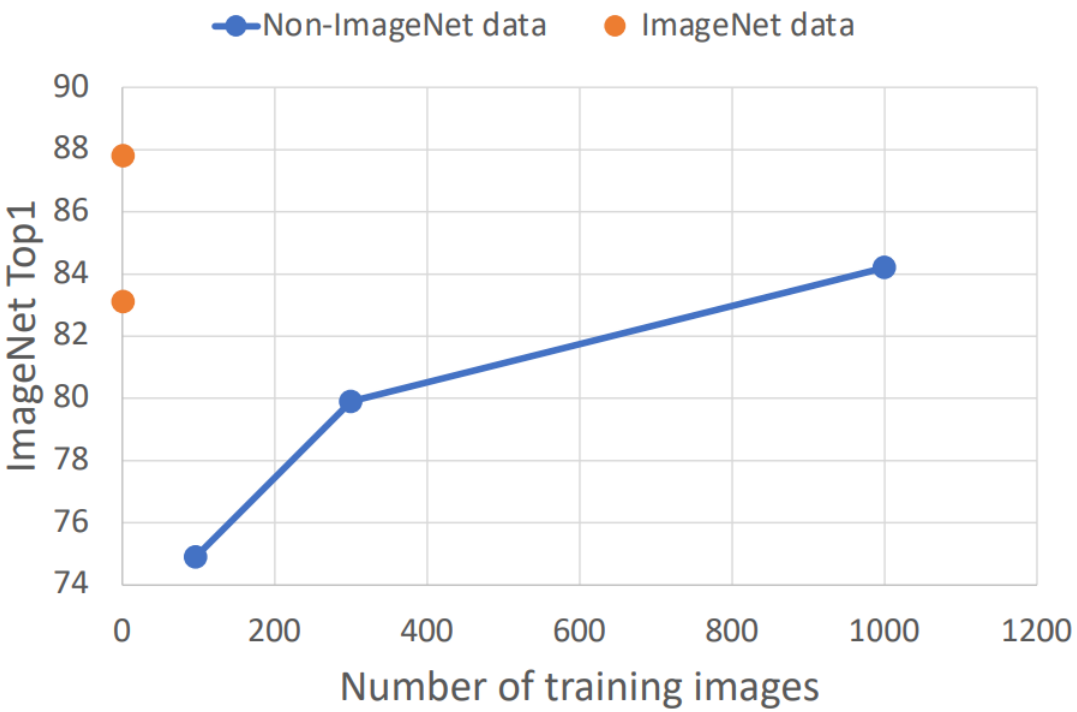
A rant about data…

The motivation of SSL is scaling to large data that can't be labeled



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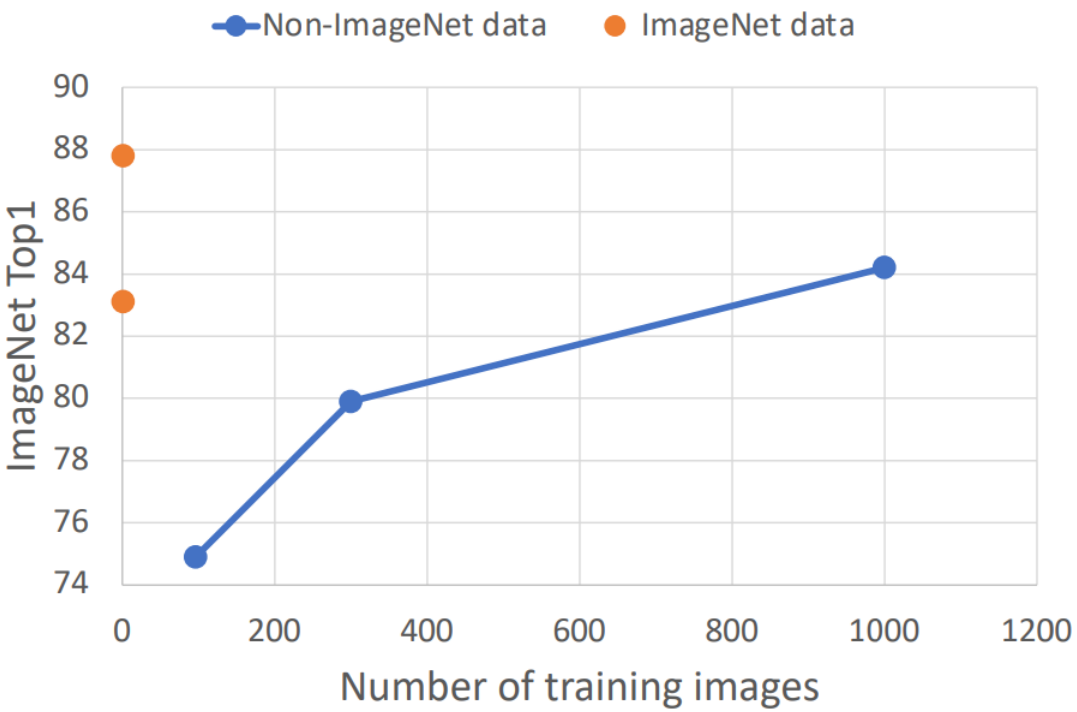
Most papers pretrain on (unlabeled) ImageNet, then evaluate on ImageNet!

Unlabeled ImageNet is still curated: single object per image, balanced classes



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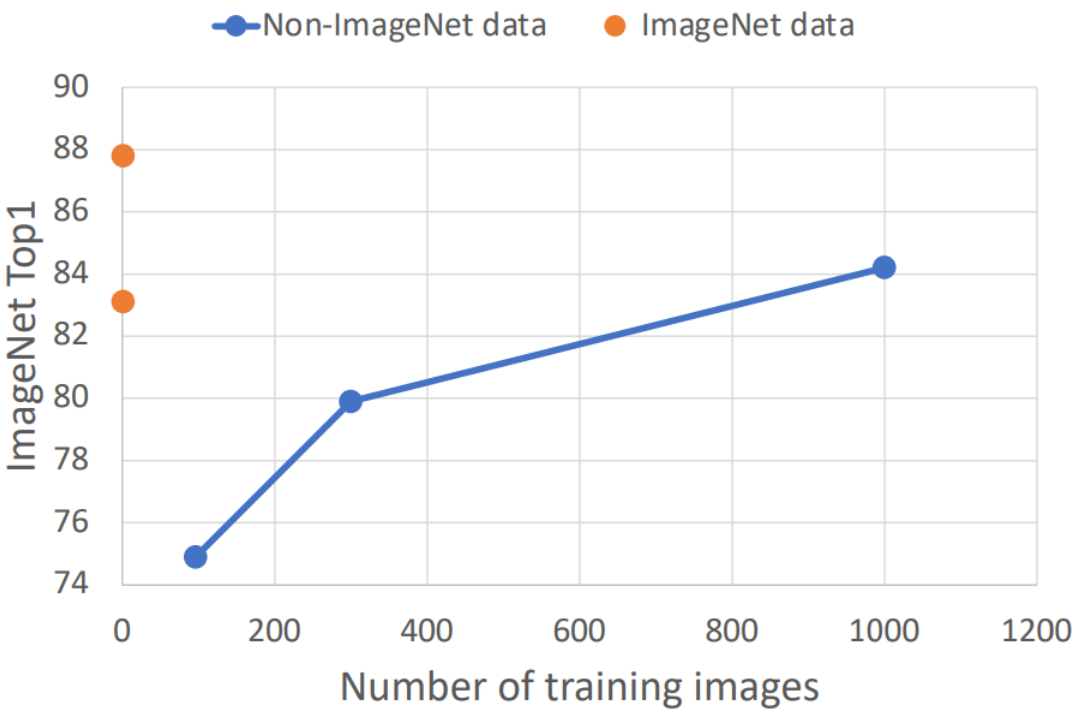

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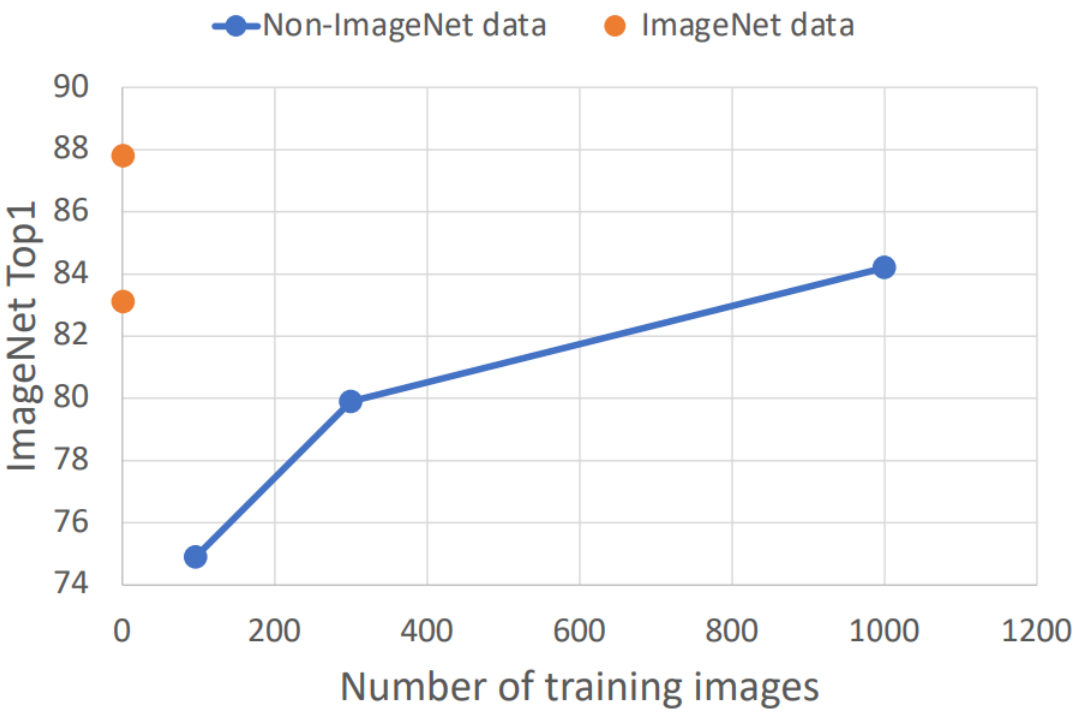
Idea: What if we go beyond isolated images?



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Multimodal Self-Supervised Learning

Don't learn from isolated images – take images together with some **context**

Video: Image together with adjacent video frames

Agrawal et al, "Learning to See by Moving", ICCV 2015
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Why Language?

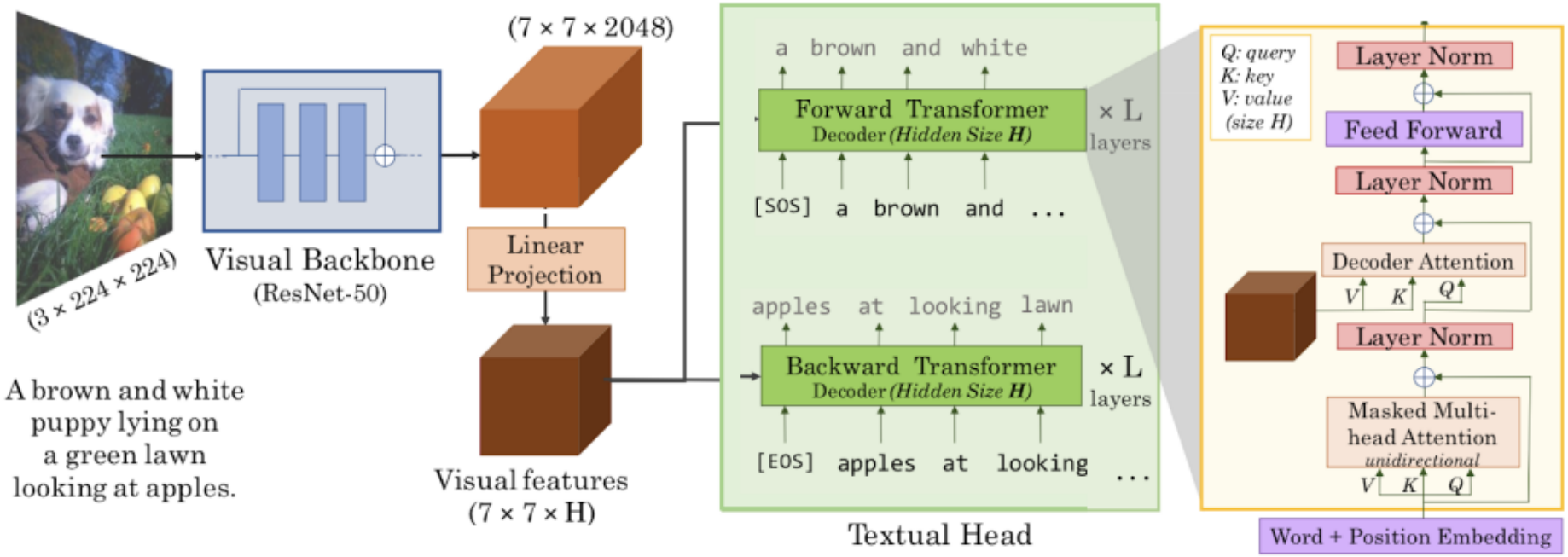
Large dataset of
(image, caption)

a dog with his head out the window of the car

a black and orange cat is resting on a keyboard and yellow back scratcher

1. **Semantic density**: Just a few words give rich information
2. **Universality**: Language can describe any concept
3. **Scalability**: Non-experts can easily caption images; data can also be collected from the web at scale

Generating Captions



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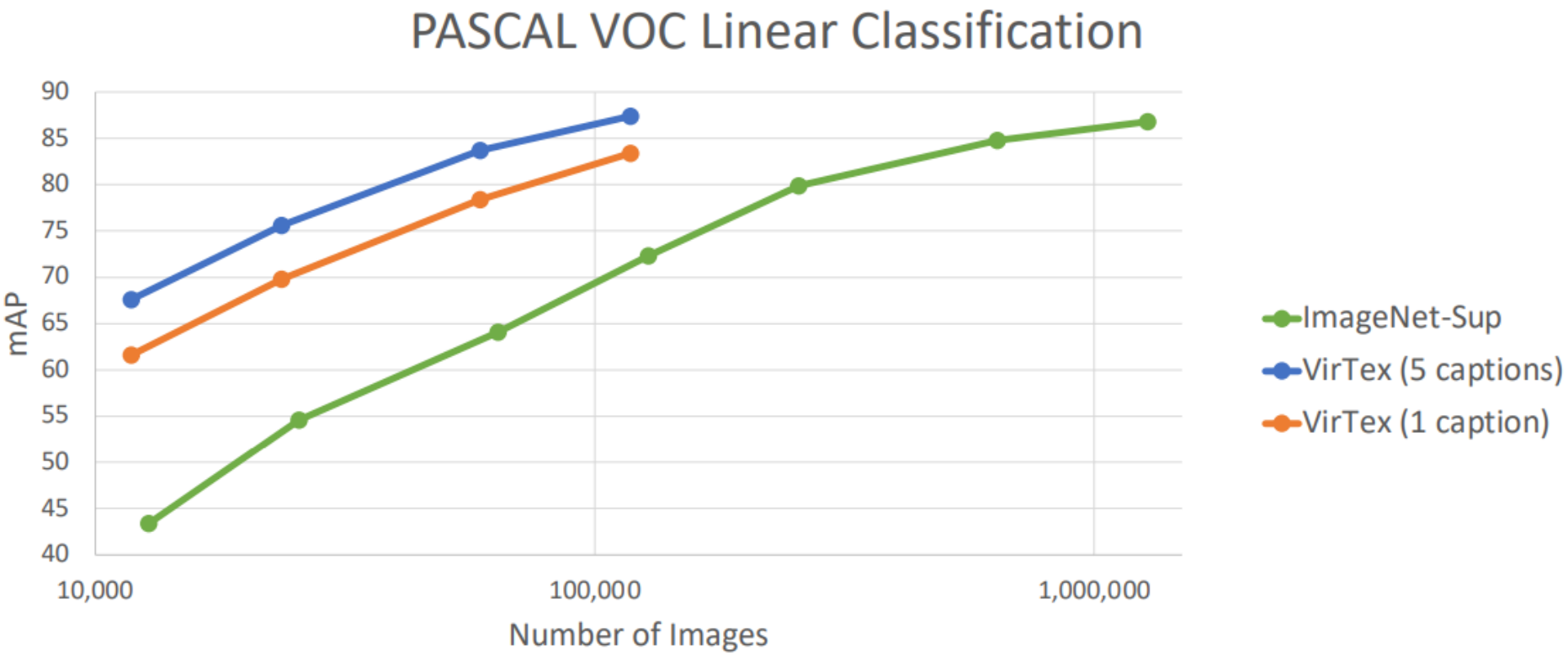

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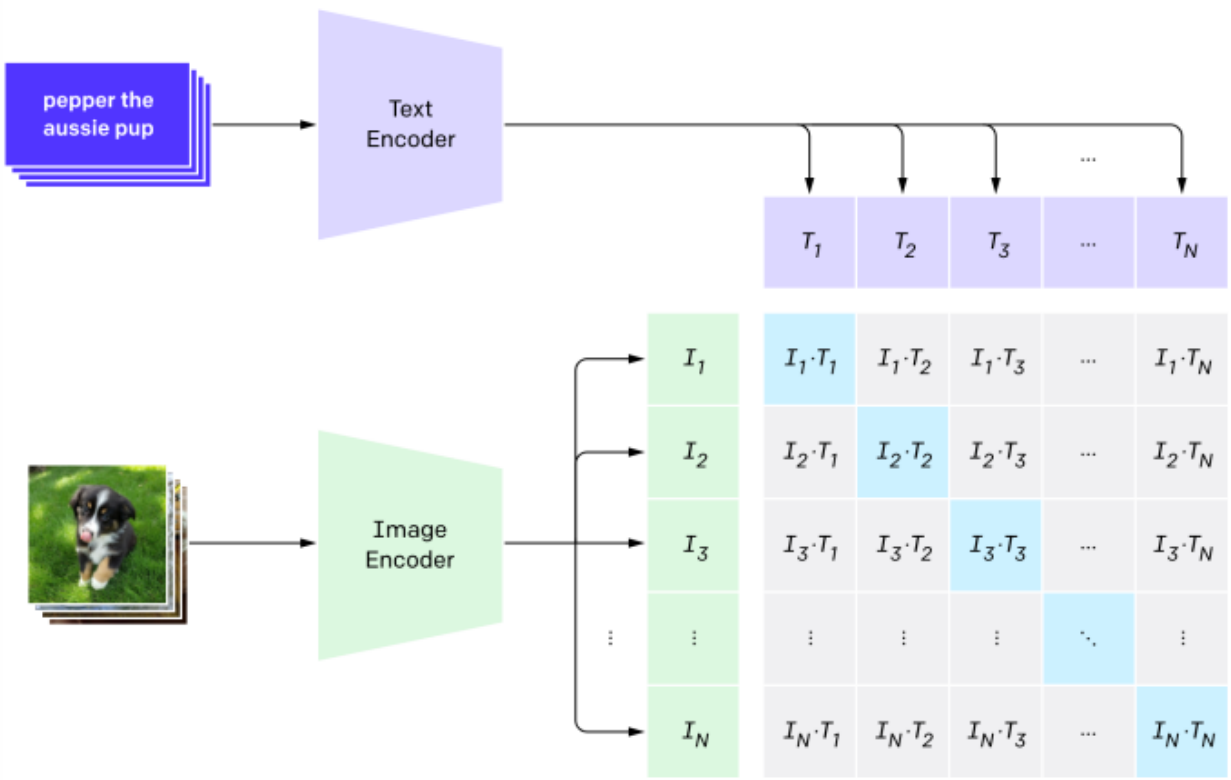
Matching Images and Text



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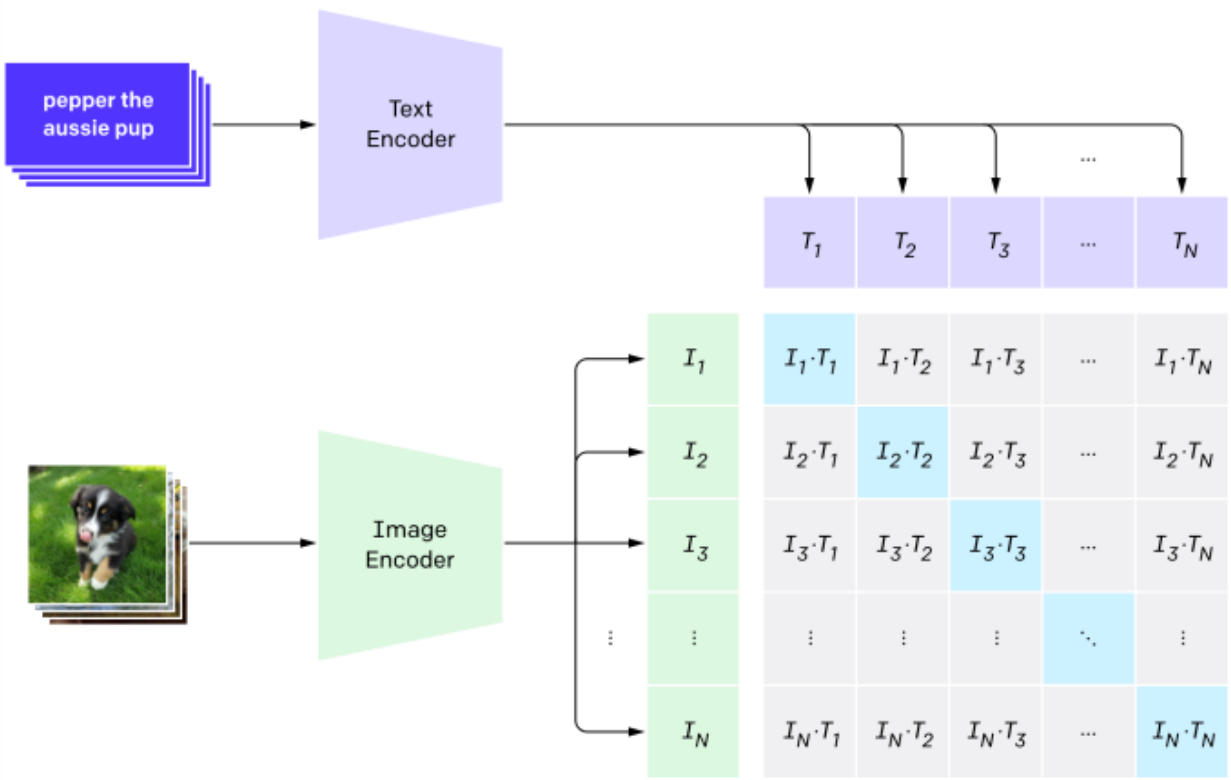

Contrastive loss: Each image predicts which caption matches

Large-scale training on 400M (image, text) pairs from the internet

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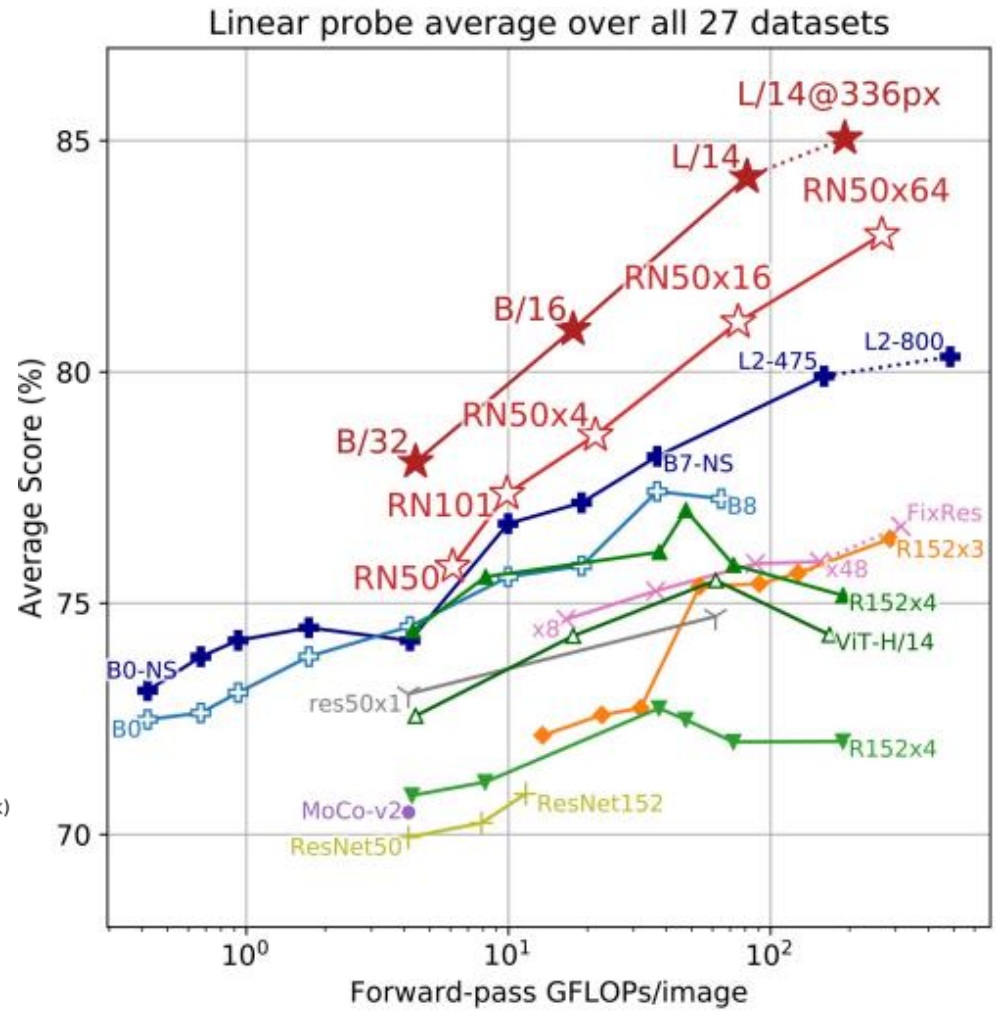
Matching Images and Text: CLIP

Very strong performance on many downstream vision problems!

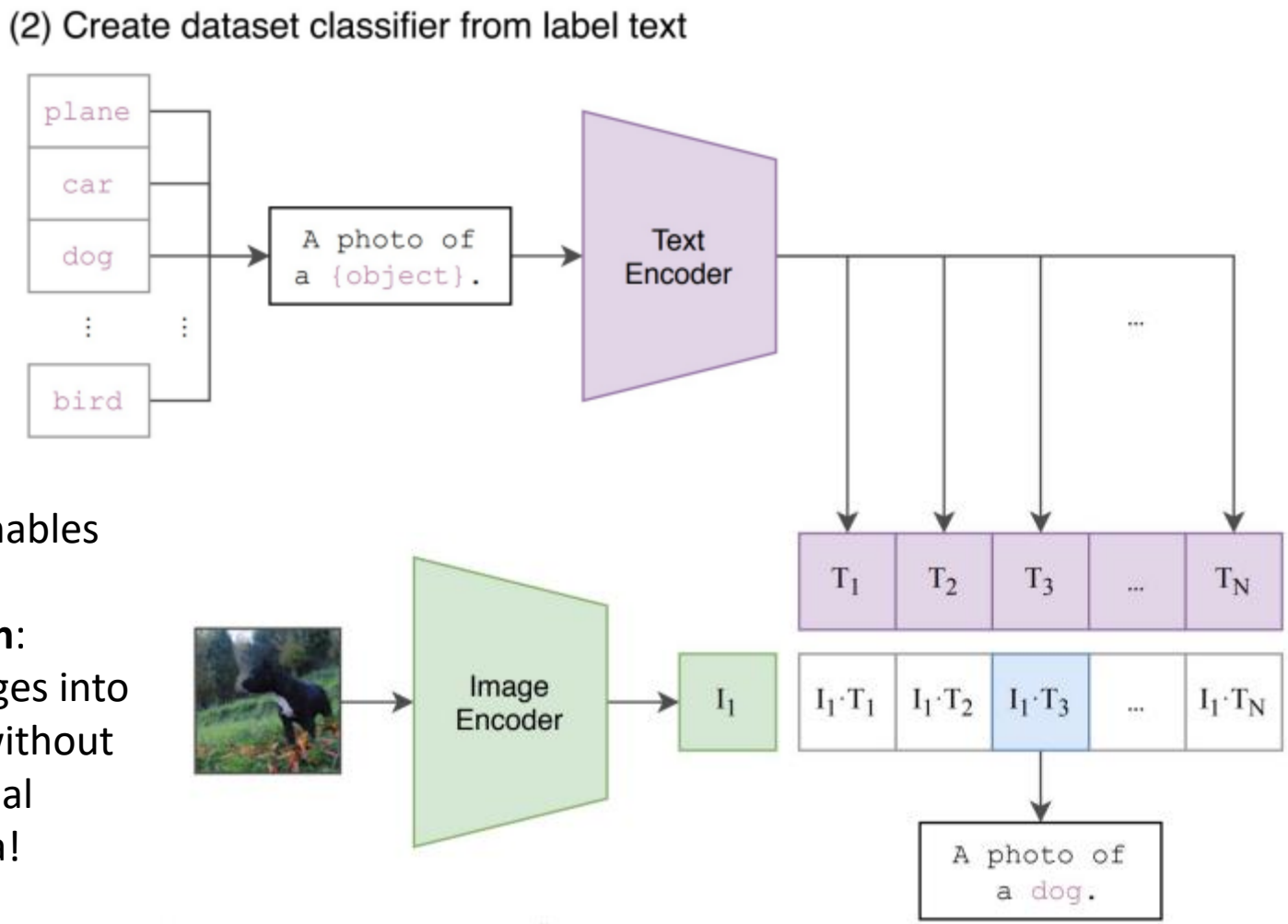
Performance continues to improve with larger models



- CLIP-ViT
- CLIP-ResNet
- EfficientNet-NoisyStudent
- EfficientNet
- Instagram-pretrained
- SimCLRv2
- BYOL
- MoCo
- ViT (ImageNet-21k)
- BiT-M
- BiT-S
- ResNet

Radford et al, “Learning Transferable Visual Models form Natural Language Supervision”, ICML 2021

CLIP: Zero-Shot Classification



Language enables **zero-shot classification**: Classify images into categories without any additional training data!

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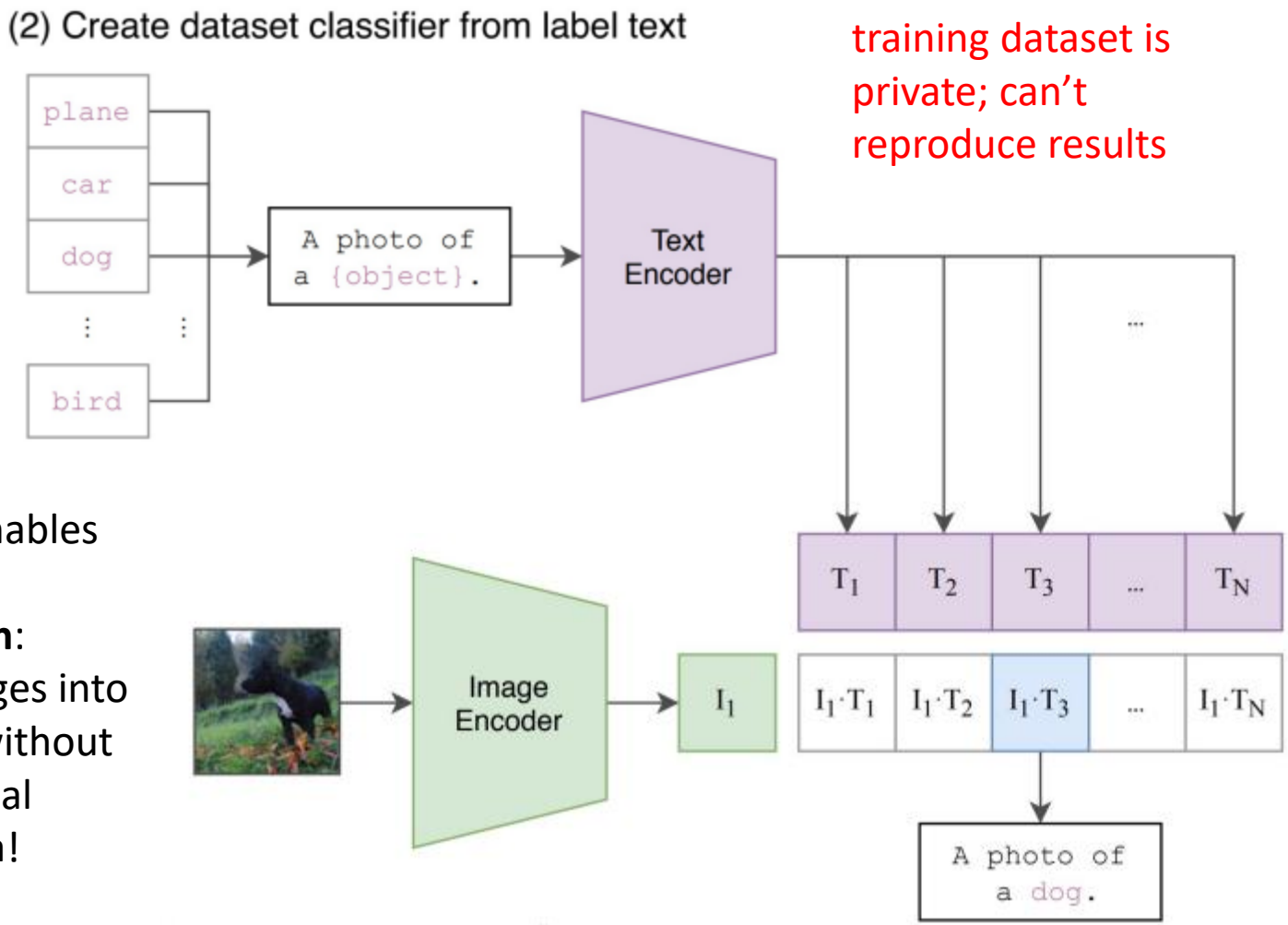
CLIP: Zero-Shot Classification

Problem: CLIP training dataset is private; can't reproduce results



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RedCaps: Images and Captions from Reddit

r/birdpics: male northern cardinal

r/crafts: my mom tied this mouse

r/itookapicture: itap of the taj mahal

r/perfectfit: this lemon in my drink

r/shiba: mlem!

Data from 350 manually-chosen subreddits
12M high-quality (image, caption) pairs

Desai, Kaul, Aysola, and Johnson, NeurIPS Datasets & Benchmarks Track, 2021

Summary

- Self-Supervised Learning (SSL) aims to scale up to larger datasets without human annotation
- First train for a **pretext** task, then **transfer** to **downstream** tasks
- Many pretext tasks: context prediction, jigsaw, colorization, clustering, rotation
- SSL has been wildly successful for language
- Intense research on SSL in vision; current best are contrastive, masked autoencoding
- Multimodal SSL uses images together with additional context
- Multimodal SSL with vision + language has been very successful; seems very promising!